Self Supervised Learning for Renal Cell Carcinoma Subtyping on Whole Slide Images

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***Abstract*—Renal cell carcinoma (RCC) subtyping is clinically critical, as treatment varies substantially across histological variants. However, diagnostic ambiguity persists, particularly for oncocytic tumors, making reliable labels scarce and thus constraining supervised learning. We benchmark four self-supervised learning (SSL) paradigms (MoCo v3, DINO v3, iBOT, I-JEPA) on 168,000 H&E patches extracted from 196 WSIs (88 patients). Beyond accuracy, we assess probabilistic reliability via calibration error analysis and selective prediction, and interpretability via attention rollout and PLIP-based concept grounding. MoCo v3 achieves the strongest patch-level performance among the SSL methods (Macro-F1 = 0.736), while patient-level findings remain preliminary due to the small held-out cohort; interpretability analyses further provide qualitative evidence that the learned representations align with morphologically plausible regions and RCC-related concepts, supporting the potential of SSL for RCC subtyping under limited supervision.**

I. Introduction

Renal cell carcinoma (RCC) accounts for approximately 90% of kidney malignancies, representing a leading cause of cancer-related mortality with over 400,000 new cases and 175,000 deaths annually worldwide [1]. Accurate histological subtyping is clinically essential because treatment strategies vary substantially: clear cell RCC typically responds to VEGF inhibitors, whereas papillary and chromophobe subtypes exhibit distinct molecular profiles and therapeutic sensitivities [2]. Prognosis, meanwhile, remains strongly stage-dependent, with 5-year survival dropping from 90% for localized disease to 19% for distant-stage cases in the United States [3]. Despite advances in immunohistochemistry and molecular profiling, morphological assessment on hematoxylin and eosin (H&E) stained slides remains the diagnostic cornerstone. However, RCC subtyping is complicated by substantial morphological heterogeneity, particularly among oncocytic tumors: distinguishing chromophobe RCC (malignant) from renal oncocytoma (benign) remains exceptionally challenging due to overlapping cytoplasmic and architectural features [4].

Digital pathology enables large-scale computational analysis of whole-slide images (WSIs) [5, 6], yet supervised deep learning faces a critical constraint: labeling a single WSI at the patch level can require 20–40 hours of specialized pathologist time [7], limiting curated datasets to hundreds of slides, far below the scale needed for robust generalization. Weakly supervised methods using slide-level labels partially alleviate this annotation burden [8, 9], but remain dependent on sizable labeled cohorts and vulnerable to domain shift when training data are limited.

Self-supervised learning (SSL) leverages large collections of unlabeled histopathology images to learn transferable representations, reducing reliance on task-specific labels during downstream adaptation [10]; here we consider four representative paradigms: contrastive learning (MoCo v3) [11], self-distillation (DINO v3) [12], masked token prediction (iBOT) [13], and joint-embedding predictive learning (I-JEPA) [14]. Systematic comparisons of these paradigms on histopathology remain limited, and prior work has focused primarily on accuracy while neglecting reliability and interpretability, critical requirements for clinical deployment.

To better reflect the complexities of clinical workflows, we evaluate these models through three orthogonal lenses. First, we assess *calibration* via Expected Calibration Error (ECE), quantifying the alignment between predicted confidence and empirical accuracy [15]. Second, we assess *selective prediction* via risk–coverage curves, characterizing the trade-off between model abstention and error reduction [16]. Third, we evaluate *interpretability* using transformer attention rollout and PLIP-based concept grounding [17] to probe whether model representations align with morphologically plausible regions and RCC-related concepts.

A. Contributions

We provide:

(i) a reproducible pipeline for WSI patch extraction and large-scale SSL pretraining on RCC histopathology;
(ii) a comparative evaluation of four SSL families on RCC subtype classification, assessed at both the patch level via linear probing and the patient level via probability averaging;
(iii) reliability assessment via calibration metrics and selective prediction;
(iv) two complementary interpretability analyses using Attention Rollout and PLIP-based concept scoring.

TABLE I
COMPARISON OF RELATED WORK IN RCC COMPUTATIONAL PATHOLOGY. OUR WORK COMBINES SYSTEMATIC SSL PARADIGM COMPARISON WITH RELIABILITY AND CONCEPT-LEVEL INTERPRETABILITY ASSESSMENT (CALIBRATION, SELECTIVE PREDICTION, PLIP).

Reference	Architecture	Dataset	XAI
Tabibu et al. [18]	CNN (ResNet)	TCGA (650 WSIs)	No
Zhu et al. [19]	CNN	Private + Multi-center	No
Campanella et al. [8]	MIL (ResNet + Attention)	Multi-cancer WSIs	No
Lu et al. [9]	MIL + SSL features	WSI (Multiple cancers)	No
Li et al. [20]	Dual-Stream MIL + SSL	TCGA	No
Chanchal et al. [21]	CNN (RCCGNet with SCR)	Private (722 WSIs)	Spatial attribution only
Zheng et al. [22]	CNN + MIL (SSL-CLAM)	TCGA (708) + CPTAC (445)	Attention heatmaps
Ciga et al. [10]	CNN (SSL contrastive)	Generic multi-organ H&E	No
Feng et al. [23]	SSL masked multi-stage	Pathology WSIs	No
Ours	**ViT-S (4 SSL paradigms)**	**Private (196 WSIs, 168k patches)**	**Attention Rollout + PLIP**

II. RELATED WORK

A. SSL Learning for Vision

Current SSL research for Vision Transformers is driven by four primary paradigms, each employing a distinct training objective. Contrastive learning, exemplified by MoCo v3 [11], focuses on distinguishing global image views. Self-distillation, such as DINO v3 [12], learns representations by matching teacher-student outputs without explicit negative samples. Masked token prediction, as implemented in iBOT [13], combines local patch reconstruction with global semantics. Finally, latent predictive learning (I-JEPA) [14] discards pixel-level details to predict abstract representations of masked regions. While these methods have demonstrated strong performance in natural image analysis, their relative effectiveness in capturing the subtle morphological nuances required for histological subtyping remains underexplored.

B. SSL for Computational Pathology

SSL leverages large unlabeled histopathology corpora to learn transferable patch encoders, reducing dependence on expert annotations [10]. Standard WSI pipelines perform SSL pretraining at the patch level and subsequently aggregate representations for slide-level or patient-level prediction.

Recent work has adapted masking-based SSL to capture fine-grained morphology and multi-scale tissue structure [23].

In kidney cancer specifically, initial studies have combined SSL features with weakly supervised frameworks [22], yet systematic comparisons across modern SSL paradigms on RCC cohorts remain limited.

C. RCC Digital Pathology

Deep learning for RCC analysis on WSIs started with patch-based CNNs applied to TCGA histopathology slides for multi-class subtype classification [18], and was extended to biopsy and surgical resection images to improve generalizability and explainability [19]. Subsequently, methods incorporating enhanced CNN modules and multi-scale feature integration demonstrated improved performance on RCC grading and subtype discrimination benchmarks [21, 24]. However, models trained on TCGA-derived WSIs may learn dataset-specific artifacts and acquisition site confounders rather than purely histological morphology, indicating sensitivity to scanner and institutional variability [25].

D. Reliability and Interpretability

For clinical deployment, calibration and selective prediction complement standard accuracy metrics: deep networks are often miscalibrated [15], and risk-coverage analysis enables safer human–AI collaboration through selective abstention [16]. Interpretability methods include transformer attention rollout for token-level saliency estimation [26] and vision-language models that ground visual features in semantic concepts through shared embedding spaces. Pathology-specific models such as PLIP improve alignment to histological descriptors, enabling concept-level explanations that complement spatial attribution [17].

Despite these advances, a systematic RCC benchmark that jointly evaluates modern SSL paradigms in terms of accuracy, reliability, and interpretability remains absent. This work addresses that gap (Table I).

III. MATERIALS AND METHODS

The experimental pipeline comprises four stages:

(i) WSI preprocessing and patch generation;
(ii) self-supervised pretraining of patch-level encoders;
(iii) downstream evaluation for RCC subtype classification at patch and patient level;
(iv) reliability and interpretability analyses.

A single preprocessing pipeline and fixed patient-level splits are used across all methods to support controlled comparison under shared data conditions.

A. Dataset and Preprocessing

a) Cohort: The dataset consists of a private cohort comprising 196 H&E WSIs from 88 patients across four diagnostic categories: clear cell RCC (`ccRCC`, 124 slides), papillary RCC (`pRCC`, 48 slides), renal oncocytoma (`ONCO`, 13 slides), and chromophobe RCC (`CHROMO`, 11 slides). In addition to tumor subtypes, the dataset includes non-neoplastic tissue annotated by pathologists (e.g., normal renal parenchyma, necrosis, and connective tissue); these regions are grouped into a single auxiliary class denoted `NOT_TUMOR`.

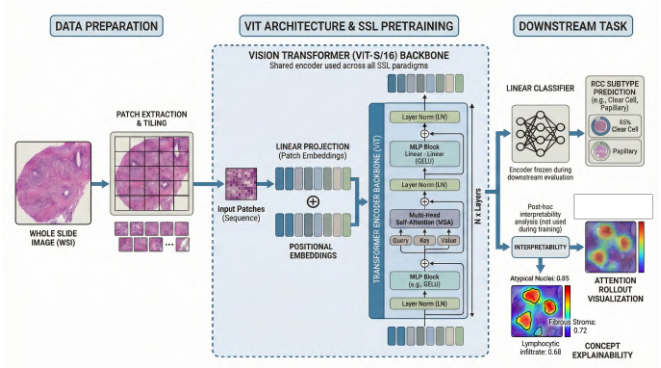

Fig. 1. Overview of the preprocessing, pretraining, and evaluation pipeline.

b) ROI sources and tissue filtering: Patch sampling is guided by WSI-level tumor regions of interest (*tumor ROIs*) annotated or selected by pathologists. For `ccRCC` and `pRCC` slides, expert-provided XML annotations are parsed to generate tumor and non-tumor masks. Conversely, for `CHROMO` and `ONCO` slides, pre-annotated tumor ROI masks delineating tumor regions were already provided within the dataset and utilized directly for sampling.

c) Patch extraction: Patches of size 224×224 pixels are extracted at 0.50 μm/px (20$\times$ equivalent). Candidate locations are sampled within tumor ROI masks, ensuring that extracted patches correspond to diagnostically meaningful tissue regions.

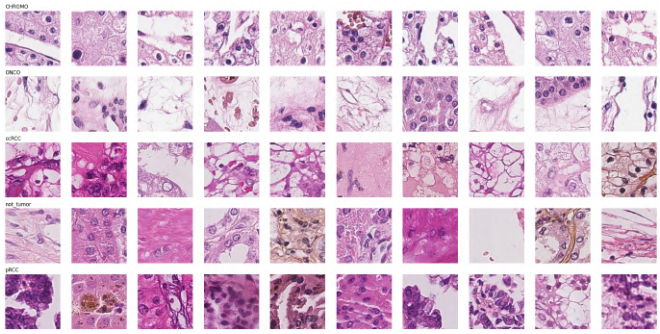

Fig. 2. Representative 224×224 px patches for each class after tumor ROI sampling and tissue filtering.

d) Splits and class balancing: Splits are defined at the patient level to prevent information leakage, with stratification across diagnostic classes. During training, class imbalance is mitigated by capping per-patient patch contributions and equalizing class frequencies to the minimum class count; non-tumor patches are sampled to match per-patient tumor counts. Validation and test sets are left unaltered.

e) Storage: Patches are serialized into compressed WebDataset shards for high-throughput streaming during training. Each patch is stored with lightweight metadata: class label, patient identifier, spatial coordinates, and source type.

B. Self-Supervised Pretraining

a) Objectives: We instantiate four representative SSL paradigms introduced earlier and pretrain them under a unified experimental protocol. All methods are trained using identical preprocessing pipelines, data splits, and backbone architectures to ensure controlled comparison. Differences between models arise primarily from their SSL objectives (contrastive, self-distillation, masked token prediction with distillation, or joint-embedding predictive learning) together with their associated method-specific training pipelines, such as view generation, multi-crop strategy, masking. The underlying patch corpus, patient splits, backbone architecture, and base preprocessing are kept consistent across experiments.

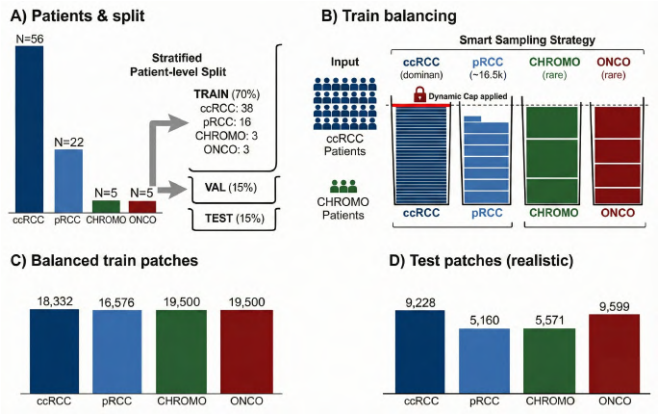

Fig. 3. Class-balanced patch distribution across RCC subtypes and train/validation/test splits.

b) Backbone and optimization: All models use a ViT-S/16 backbone (16×16 pixel patches, 224 px input). Training uses AdamW with cosine learning-rate scheduling and automatic mixed precision. Standard augmentations include random resized crops, horizontal and vertical flips, Gaussian blur, color jitter, and grayscale conversion. Histology-specific augmentations (HED jitter, Macenko stain normalization) are explored in ablation studies (Appendix B).

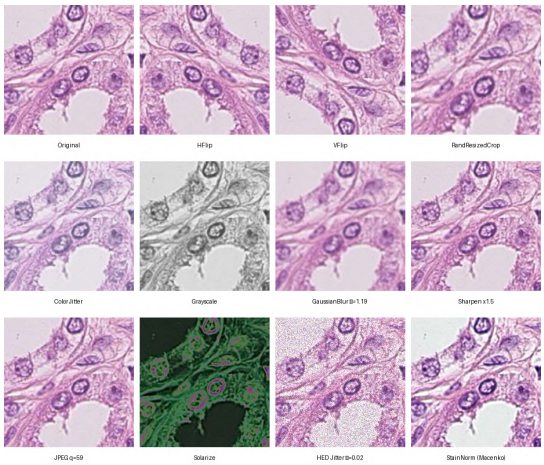

Fig. 4. Augmentation examples applied to a single patch.

C. Evaluation Protocol

a) Patch-level classification: Representation quality is assessed by freezing SSL encoders and training a linear classifier (linear probing) on labeled patches from the training split. Hyperparameters are selected on the validation split; final

results are reported on the test split across five classes (ccRCC, pRCC, CHROMO, ONCO, NOT_TUMOR).

b) Metrics: We report accuracy, Macro-F1 (MF1), balanced accuracy (BAcc), one-vs-rest AUROC, PR-AUC, and the minimum recall across tumor classes, MinRec(T) (excluding NOT_TUMOR). MinRec(T) surfaces worst-case per-class sensitivity, which aggregate ranking metrics such as AUROC can mask in safety-critical settings.

c) Supervised baselines: Two supervised baselines are included for reference, both trained on the same RCC patch dataset used throughout this study (i.e., the 224×224 H&E patches extracted from our annotated WSIs under the same train/validation/test splits).

(i) a ResNet-50 trained from scratch on the labeled training split (*Supervised Scratch*);
(ii) a ResNet-50 initialized with ImageNet pretrained weights and fine-tuned on the same training data (*Supervised Transfer*).

ResNet-50 was selected because it is a strong and widely used CNN reference architecture in computational pathology, providing a conventional supervised baseline against which ViT-based SSL models can be compared. Its model capacity is of the same order of magnitude as ViT-S/16, making the comparison informative at the task level. These baselines share the same splits, preprocessing, and evaluation endpoints as the SSL models, but differ in backbone architecture and downstream adaptation protocol. We therefore report them as task-level reference baselines rather than as an architecture-matched isolation of supervision alone.

d) Patient-level aggregation: Patient-level predictions are obtained by averaging patch-level class probabilities across all patches belonging to a given patient, followed by renormalization over tumor classes (excluding NOT_TUMOR). Majority voting is reported as an alternative baseline. Patient-level evaluation is computed on $N_{\text{pat}} = 12$ held-out tumor patients.

D. Reliability

a) Calibration: We quantify calibration using Expected Calibration Error (ECE) [15], computed with 15 equal-width bins as the weighted average of absolute differences between bin confidence and bin accuracy. Confidence is defined as the maximum softmax probability over classes.

b) Selective prediction: We assess confidence-based abstention via risk–coverage curves [16], which characterize the trade-off between error rate and retained sample fraction under varying confidence thresholds. Abstaining on low-confidence predictions allows models to reduce error at the cost of coverage, supporting safer human–AI collaboration in diagnostic workflows.

E. Interpretability

a) Attention rollout: For transformer-based models, spatial attribution is computed via attention rollout [26], which aggregates attention weights across layers to approximate the contribution of each image region to the final representation. In our implementation, the rollout map is thresholded (top-quantile) to obtain a binary high-attention mask. Regions with high rollout scores are localized and converted into bounding-box contours that define attention-derived regions of interest (ROI) within each patch (i.e., model-derived ROIs, distinct from the WSI-level tumor ROIs used for sampling). These contours are overlaid on the original patch, producing a visualization that highlights the spatial areas most attended by the SSL encoder. The resulting image is then used as input to PLIP for concept retrieval. Qualitative examples are reported in Fig. 8 and Appendix F.

b) Concept ontology construction: We first assemble a concept ontology of 64 visually grounded H&E descriptors designed for 224×224 tiles. Candidate concepts were drafted via an LLM-assisted workflow using *GPT-5 Thinking* (OpenAI) to propose concepts and draft definitions, and we then manually curated them into concise tile-level morphology descriptors while preserving diagnostic relevance [27]. Definitions draw on standard RCC morphology references and normal kidney histology sources [2, 28–36]. The calibrated final concept bank derived from this ontology is reported in Appendix D.

c) Semantic concept validation: To validate whether model predictions align with clinically meaningful morphology, we use PLIP as an independent, external concept scorer [17]. For each selected qualitative case, we score PLIP concepts on the attention-guided patch view and report the top-ranked concepts alongside the attention overlays. Image embeddings are computed with the PLIP image encoder; in parallel, the final concept bank is encoded with the PLIP text encoder. Patch–concept alignment is quantified via scaled cosine similarity (CLIP-like logits; we report $100\times$ cosine similarity). To derive the class-specific final concept bank, we run a separate concept calibration and deep-validation stage on the training and validation splits, scoring the full concept ontology and estimating per-concept discriminativeness (e.g., one-vs-rest AUROC) and confounding behavior. The resulting final concept bank is then fixed and used for test-time XAI (Appendix E).

F. Reproducibility

All experiments were conducted on NVIDIA A40 GPUs (48 GB VRAM) using PyTorch. Data loading was implemented using WebDataset sharded archives to enable efficient streaming and GPU-aware prefetching. All experiments use fixed random seeds (seed=42) and deterministic algorithms where available to support reproducibility.

Training configurations are fully specified via YAML files that define hyperparameters, augmentation policies, optimization schedules, and evaluation protocols. The codebase follows a modular architecture with clear separation between preprocessing, SSL pretraining, linear probing, and patient-level aggregation.

Installation instructions and execution guidelines are documented in the GitHub repository to support reproduction of the workflow, subject to dataset availability. The codebase is

modular and can be adapted to other histopathology datasets without changing its overall structure.

IV. Results

We evaluate all models on the held-out test split at both patch level (five-way classification: `ccRCC`, `pRCC`, `CHROMO`, `ONCO`, `NOT_TUMOR`) and patient level using tumor-only aggregation (`NOT_TUMOR` excluded).

A. Performance Comparison and Worst-Case Sensitivity

Tables II and III report the full metric set for patch and patient-level evaluation. Rows report the best ablation per model family selected by patch-level Macro-F1 (full ablation results in Appendix B).

TABLE II
PATCH-LEVEL RESULTS. ROWS REPORT THE BEST ABLATION PER MODEL FAMILY SELECTED BY PATCH-LEVEL MACRO-F1.

Model	Acc	MF1	BAcc	AUROC	PR-AUC	MinRec(T)
MoCo v3	0.750	0.736	0.739	0.948	0.812	0.563
DINO v3	0.607	0.615	0.632	0.883	0.643	0.486
iBOT	0.618	0.529	0.596	0.919	0.704	0.040
I-JEPA	0.490	0.464	0.551	0.864	0.590	0.159
Sup. Scratch	0.813	0.819	0.819	0.947	0.852	0.555
Sup. Transfer	0.851	0.857	0.863	0.963	0.876	0.742

At patch level, *Supervised Transfer* yields the most balanced performance across metrics, followed by *Supervised Scratch*. Among SSL methods, MoCo v3 performs best. Notably, several SSL methods show severe failures on specific tumor classes despite acceptable aggregate metrics: iBOT achieves MinRec(T)=0.040, indicating very low sensitivity to ONCO, while I-JEPA reaches only 0.159 minimum recall. These worst-case sensitivities highlight that ranking-based metrics (AUROC) can mask clinically problematic blind spots.

a) Ablation highlights.: We analyze per-family sensitivity to key hyperparameters and loss components. For MoCo v3, the best-performing configuration in our runs is `abl03` (τ=0.4), suggesting that the contrastive temperature is a relevant tuning knob under our fixed SSL batch size of 64 (Table VII). For DINO v3, comparing `abl06` ($\lambda_{\text{iBOT}} = 0$, $\lambda_{\text{KoLeo}} = 0.1$, $\lambda_{\text{GRAM}} = 1.0$) against `abl02` ($\lambda_{\text{iBOT}} = 0$, $\lambda_{\text{KoLeo}} = 0$, $\lambda_{\text{GRAM}} = 0$; Table IX) suggests that auxiliary regularization partially prevents full collapse, but does not recover strong downstream features without the patch-level iBOT term. In particular, `abl06` improves over `abl02` (Macro-F1 0.44 vs. 0.22), yet remains the weakest non-degenerate DINO v3 configuration and reaches only MinRec(T)= 0.219. This indicates that KoLeo/GRAM help stabilize training, whereas the iBOT patch objective is the main driver of representation quality in our patch-level linear-probe setting. For iBOT, MinRec(T) remains low across ablations (0.020–0.145) and is 0.040 even for the best Macro-F1 run (`abl04`), highlighting a persistent worst-case sensitivity limitation. For I-JEPA, several configurations exhibit worst-case recall collapse (MinRec(T)=0.000 in `abl05`/`abl08`/`abl10`/`abl12`), underscoring sensitivity to architectural and masking choices beyond average AUROC. Full ablation tables and per-run results are provided in Appendix B.

TABLE III
PATIENT-LEVEL RESULTS.

Model	Acc	MF1	AUROC	PR-AUC	MinRec(T)
MoCo v3	0.917	0.945	0.984	0.975	0.857
DINO v3	0.750	0.786	0.920	0.898	0.571
iBOT	0.833	0.612	0.967	0.943	0.000
I-JEPA	0.417	0.354	0.829	0.791	0.000
Sup. Scratch	0.833	0.881	0.960	0.951	0.667
Sup. Transfer	0.833	0.881	0.984	0.975	0.429

At patient level, MoCo v3 attains the highest Macro-F1 (0.945), matching or exceeding supervised baselines (0.881) after probability-averaging aggregation. Given the limited test cohort ($N = 12$), these findings remain preliminary and require validation on larger external datasets. Furthermore, the discrepancy between AUROC and Macro-F1 is particularly evident for methods like iBOT, where high AUROC (0.967) coexists with MinRec(T)=0.000.

B. Error Analysis

We analyze the dominant confusions for MoCo v3 at both patch and patient level.

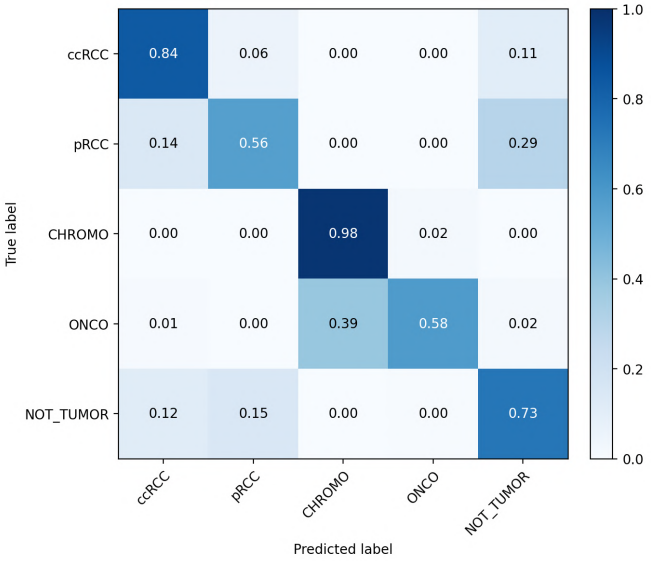

Fig. 5. Patch-level confusion matrix for MoCo v3.

Misclassifications primarily occur between morphologically related tumor subtypes, particularly **`ONCO` vs. `CHROMO` (39% of `ONCO` patches misclassified as `CHROMO`)** and **`ccRCC` vs. `pRCC` (`ccRCC`→`pRCC`: 6%; `pRCC`→`ccRCC`: 14%)**. These confusions align with known diagnostic challenges in oncocytic tumor differentiation [4], where even expert pathologists exhibit inter-observer disagreement ($\kappa = 0.32$). Notably, `NOT_TUMOR` remains comparatively well separated (recall $\approx$ 0.73), indicating that the primary challenge lies in subtype discrimination rather than tumor detection.

At patient level (Fig. 6), aggregation reduces some patch-level noise but residual confusions persist.

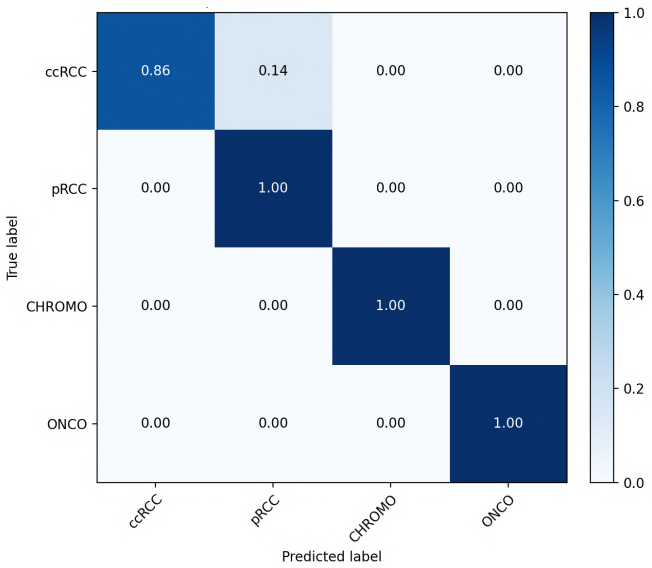

Fig. 6. Patient-level confusion matrix for MoCo v3 using tumor-only aggregation.

C. Calibration Analysis

We quantify calibration using Expected Calibration Error (ECE) with 15 bins, measuring the alignment between predicted confidence and empirical accuracy. At patch level, all models exhibit miscalibration to varying degrees (Table IV). Among the SSL methods, MoCo v3 achieves the lowest ECE (0.234), whereas I-JEPA shows the highest miscalibration (0.482). The supervised reference baselines yield lower ECE overall (*Supervised Transfer*: 0.124; *Supervised Scratch*: 0.122).

TABLE IV
PATCH-LEVEL CALIBRATION (ECE, 15 BINS) ON THE TEST SET.

Model	ECE
MoCo v3	0.234
DINO v3	0.372
iBOT	0.356
I-JEPA	0.482
Sup. Scratch	0.122
Sup. Transfer	0.124

These findings indicate that, in our patch-level evaluation setup, SSL-derived confidence estimates require recalibration before deployment in safety-critical diagnostic settings. The possible role of backbone and adaptation differences, together with post-hoc calibration strategies such as temperature scaling, is discussed in Section V.

D. Selective Prediction

We assess confidence-based abstention through risk–coverage analysis at patch level for MoCo v3 (Fig. 7). Empirical risk decreases as coverage is reduced, indicating that model confidence provides informative uncertainty estimates. These trends suggest that selective abstention can meaningfully improve reliability, although calibration limitations remain.

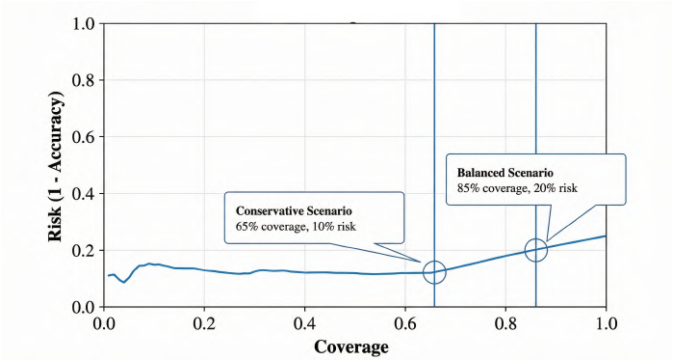

Fig. 7. Risk–coverage curve for MoCo v3 on patch-level predictions.

E. Interpretability Analysis

We evaluate interpretability through two complementary approaches: transformer attention rollout for spatial attribution and PLIP-based concept grounding for semantic validation.

Spatial attribution. Attention rollout suggests that MoCo v3 often focuses on morphologically plausible tissue regions rather than background or obvious staining artifacts. Qualitative examples frequently exhibit concentrated attention over features consistent with clear-cell cytoplasm (`ccRCC`), papillary cores (`pRCC`), or chromophobe halos (`CHROMO`).

Concept grounding. PLIP-based analysis provides a concept-level semantic audit of learned representations (Fig. 8 and Appendix F). Scores are PLIP CLIP-like logits, i.e., scaled cosine similarities between normalized image and text embeddings using the model's learned logit scale; relative rankings are the main quantity of interest. We report concept IDs (e.g., C028) from the Final Concept Bank (Appendix Table XV).

Across qualitative examples, PLIP concepts often agree with the metadata labels. In the MoCo example (T=`CHROMO`, P=`CHROMO`), top-ranked concepts are chromophobe-associated, including perinuclear halos (C028), oncocytic variant (C027), pale reticulated (C026), and microvesicular cytoplasm (C029). The DINO example illustrates ONCO–CHROMO ambiguity (T=`ONCO`, P=`CHROMO`) with a mixture of oncocytic/chromophobe concepts such as perinuclear halos (C028), microvesicular cytoplasm (C029), and round nuclei no halos (C036). The iBOT example highlights a failure mode: despite a non-tumor ground truth (T=`NOT_TUMOR`), the model predicts pRCC (P=`pRCC`) while PLIP top concepts emphasize non-tumor/context cues (e.g., lymphocytes C024, thick vessels C025, distal collecting C052, peritumoral kidney C021). Finally, the I-JEPA example (T=`ONCO`, P=`ONCO`) ranks oncocytoma-associated concepts such as polygonal cells prominent nucleoli (C041), oncocytic cytoplasm (C040), nested trabecular (C038), and tubulocystic (C042).

Quantitative concept audits (Appendix E) show that several top-ranked concepts per class achieve $\text{AUROC}_{\text{OvR}} > 0.6$ for putative primary diagnostic features, providing supporting evidence that the learned representations retain RCC-relevant morphological information despite the absence of concept-level supervision and the limited alignment between

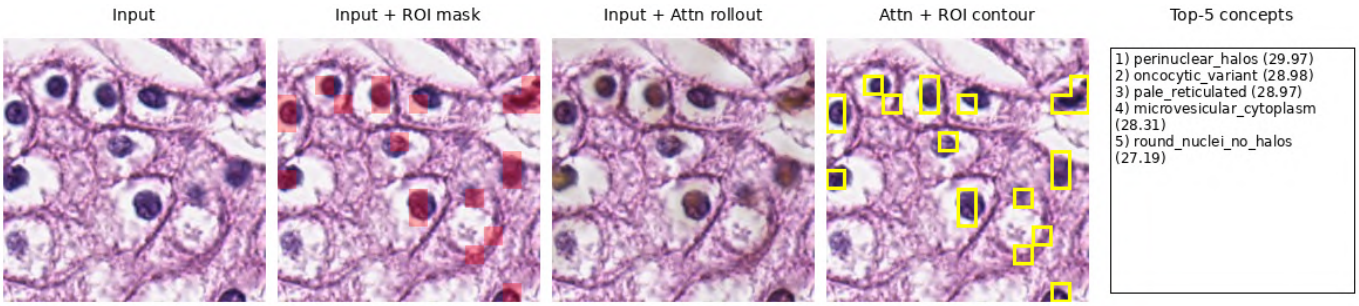

Fig. 8. PLIP-based concept grounding for MoCo v3. **Left to right:** (a) Original H&E patch (224×224 px), (b) patch + high-attention mask (thresholded rollout map; top-quantile tokens), (c) attention rollout heatmap showing spatial attribution, (d) attention-guided PLIP input (patch + attention-derived ROI contours; yellow; derived from the thresholded high-attention mask), (e) top-5 PLIP concept logits (scaled cosine similarity).

the generic PLIP encoder and RCC morphology.

Taken together, these analyses provide qualitative and concept-level evidence that the learned representations are at least partially aligned with RCC-relevant spatial and semantic cues, although these explanations should not be interpreted as causal evidence for model decision-making.

V. Discussion

The benchmark indicates that SSL objective choice materially shapes RCC representation quality and downstream reliability.

a) Objective alignment with histopathological variability: The superior performance of MoCo v3 suggests that instance discrimination with explicit negatives is effective at capturing local discriminative features relevant for RCC subtype separation [10, 11]. Histopathology discrimination often relies on subtle cytoplasmic texture, nuclear morphology, and micro-architecture; contrastive SSL encourages representations that preserve such fine-grained differences by directly contrasting instances. In contrast, distillation and predictive objectives that emphasize invariance and abstraction can be highly effective at scale in natural-image settings, but may be less aligned with subtle cytologic cues under limited domain-specific data [12, 14, 37]. Masked-token objectives can be strong in pathology, yet they remain sensitive to masking ratio and augmentation design; the class-wise collapse observed with iBOT in our benchmark motivates careful tuning and pathology-specific recipes [13, 23].

b) Failure modes beyond average metrics: Aggregate metrics such as Macro-F1 or AUROC can obscure severe class-specific deficiencies, whereas the minimum recall metric exposes failures at the operating point. This is particularly important for oncocytic tumors: distinguishing malignant chromophobe RCC from benign renal oncocytoma is a well-documented diagnostic challenge, with inter-observer agreement reported as low as Cohen's $\kappa = 0.32$ (95% CI 0.09–0.55) [4, 38].

c) Representation stability under aggregation: Patient-level aggregation appears to smooth patch-level noise and emphasize stable features across many local observations. The MoCo pattern is consistent with the possibility that contrastive SSL yields representations that remain useful after simple aggregation, but the held-out patient cohort is too small for strong claims.

d) Calibration and confidence estimation: The observed miscalibration indicates that predicted probabilities should not be treated as calibrated risk estimates without post-hoc correction [15]. Comparisons with supervised baselines should not be over-generalized because they also reflect differences in downstream adaptation protocol and backbone family. A cautious interpretation is that the SSL-derived patch classifiers considered here do not produce calibrated probabilities out of the box. This is plausible because self-supervised objectives optimize representation invariance or latent alignment rather than class-posterior estimation, and the linear probe has limited capacity to reshape confidence after pretraining. Risk–coverage analysis nonetheless shows that confidence retains useful ranking information, providing a practical mechanism to trade coverage for lower error and support safe human–AI collaboration when the model abstains on low-confidence cases [16]. Taken together, these findings indicate that post-hoc recalibration is likely necessary before clinical use of model probabilities.

e) Interpretability and semantic alignment: Attention rollout and PLIP-based concept grounding provide complementary views of model behavior, but should be interpreted cautiously [17, 26]. Spatial attribution can suggest attention to morphologically plausible regions, while concept similarity scores offer a concept-based audit of attention-derived ROIs against an explicit final concept bank derived from the concept ontology during calibration. Nevertheless, neither attention maps nor concept similarity implies causal reliance, and PLIP is an external model with imperfect alignment to RCC morphology. Expert-annotated concept labels are required to assess faithfulness and to quantify the extent to which explanations reflect diagnostic evidence rather than correlates.

Limitations and Future Work

This study has two primary limitations. First, patient-level evaluation is based on only $N_{\text{pat}}(T)$=12 held-out tumor patients, so these results should be interpreted as preliminary. Future work should therefore validate the findings on larger, multi-center cohorts to assess robustness to staining protocols, scanner variability, and acquisition conditions. Second, experi-

ments on larger datasets and with a broader range of backbone capacities are needed to determine whether the observed differences across SSL paradigms primarily reflect the pretraining objective or the limited data regime of this study. In addition, future work should evaluate post-hoc recalibration strategies for SSL-derived confidence scores and test whether calibration remains stable at both patch and patient level.

VI. Conclusions

We compared four representative self-supervised learning paradigms for RCC histopathology under a shared evaluation protocol.

Contrastive pretraining (MoCo v3) is the strongest SSL approach in this benchmark, which is consistent with the view that explicit instance discrimination is better aligned with fine-grained RCC morphology than the more invariant or predictive objectives considered here.

Patient-level aggregation results are encouraging, but they remain preliminary because the held-out patient cohort is small.

Probability calibration remains imperfect, and the interpretability analyses should be treated as supportive rather than causal evidence. These two issues make post-hoc recalibration and cautious explanation audits natural requirements before clinical deployment.

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APPENDIX A
IMPLEMENTED SSL ARCHITECTURES

A. Overview

This study evaluates four complementary self-supervised learning (SSL) paradigms that represent distinct inductive biases for learning visual representations without manual annotations: Momentum Contrast (MoCo v3) [11], self-distillation (DINO v3) [12], Image BERT pre-training with Online Tokenizer (iBOT) [13], and Joint Embedding Predictive Architecture (I-JEPA) [14].

All methods are implemented with the same ViT-S/16 backbone at 224×224 resolution and share the same base preprocessing. Only the method-specific view generation, masking, and branch interactions differ across paradigms.

a) Common backbone and notation.: Given an input patch image x, two stochastic augmentations t_1, t_2 produce views $v_1 = t_1(x)$ and $v_2 = t_2(x)$. A ViT encoder $f_\theta(\cdot)$ maps a view to a sequence of token embeddings, from which either the `[CLS]` token or patch tokens are used depending on the paradigm. In all four evaluated methods, a second branch (key/teacher/target, depending on the method) is stop-gradient and updated as an exponential moving average (EMA) of the student parameters:

$$\bar{\theta} \leftarrow m\,\bar{\theta} + (1 - m)\,\theta, \tag{1}$$

where $m \in [0, 1)$ is the momentum coefficient.

b) Why these paradigms.: MoCo v3 emphasizes instance discrimination via contrastive objectives and explicit negatives; DINO v3 learns by matching teacher–student distributions and is known to produce strong semantic clustering; iBOT adds masked-image modeling to distillation, encouraging token-level semantics; I-JEPA removes reconstruction and contrastive negatives entirely, learning by predicting latent targets, which can reduce reliance on low-level cues.

B. Summary Comparison of SSL Paradigms

Table V summarizes the conceptual differences between the evaluated methods, alongside the best results achieved in our runs under a unified ViT-S/16 setup.

TABLE V
COMPARATIVE SUMMARY OF THE EVALUATED SSL PARADIGMS AND THEIR BEST EXPERIMENTAL RESULTS (VIT-S/16).

Aspect	**MoCo v3**	**DINO v3**	**iBOT**	**I-JEPA**
Primary objective	Contrastive (instance discr.)	Distillation (matching)	MIM + distillation	Latent prediction
Explicit negatives	Yes (in-batch, no queue)	No	No	No
Momentum branch	Key encoder	Teacher	Teacher	Target encoder
Output type	Embedding similarity	Prob. dist. over prototypes	Token dist. (masked)	Latent vectors
Multi-crop aug.	No (standard 2-view)	Yes	No (standard 2-view)	No (context/target blocks)
Collapse avoidance	Negatives + momentum	Sinkhorn-Knopp + temperature (+ KoLeo)	centering + mask	predictor + EMA target
Experimental Results				
Patch Macro-F1	0.736	0.615	0.529	0.464
Patient Macro-F1	0.945	0.786	0.612	0.354

C. MoCo v3 (Momentum Contrast)

Principle. MoCo v3 learns representations by pulling together embeddings of two augmented views (**View 1** and **View 2**) of the same image (positive pair) while pushing them away from embeddings of other images in the current batch (negatives). As illustrated in Figure 9, it employs a dual-branch architecture.

The **Student Branch** is the **query encoder** and is updated actively via backpropagation; it consists of a **Backbone Student**, a **Projector Student**, and a **Predictor Student**. The **Teacher Branch** is the **key encoder** and consists of a **Backbone Teacher** and a **Projector Teacher**. The teacher/key weights are not updated via gradients; in our implementation both **Backbone Teacher** and **Projector Teacher** are updated through a slow **Momentum Update** (Exponential Moving Average) of the corresponding student/query modules.

Notably, MoCo v3 discards the memory queue used in previous versions, relying instead on large batch sizes to provide a sufficient number of negative samples.

a) Objective (high level).: Let **Student Output 1** (q_1) and **Student Output 2** (q_2) be the query embeddings generated by the Student/query branch for View 1 and View 2, respectively: $q_1 = f_q(v_1)$ and $q_2 = f_q(v_2)$. Let **Teacher Output 1** (k_1) and **Teacher Output 2** (k_2) be the corresponding key embeddings generated by the Teacher/key branch: $k_1 = f_k(v_1)$ and $k_2 = f_k(v_2)$. To maximize the informational yield, MoCo v3 optimizes a **Symmetric InfoNCE Loss**. For a batch of size N, the contrastive loss in one direction (e.g., matching Student Output 1 with Teacher Output 2) is formulated as:

$$\mathcal{L}_{\text{ctr}}(q_1, k_2) = -\log \frac{\exp(\text{sim}(q_1, k_2)/\tau)}{\sum_{j=1}^{N} \exp(\text{sim}(q_1, k_{2,j})/\tau)}, \tag{2}$$

where $\text{sim}(\cdot, \cdot)$ is the cosine similarity and τ is a temperature parameter.

The total objective is the symmetrized sum of both cross-view predictions:

$$\mathcal{L}_{\text{Symmetric}} = \mathcal{L}_{\text{ctr}}(q_1, k_2) + \mathcal{L}_{\text{ctr}}(q_2, k_1). \tag{3}$$

b) Practical implication.: The presence of explicit in-batch negatives typically improves separation between visually similar classes, but can also over-emphasize stain/texture cues if augmentations are not strong enough. In histopathology, careful color augmentation is therefore important to reduce reliance on superficial color shortcuts and encourage stronger use of morphology.

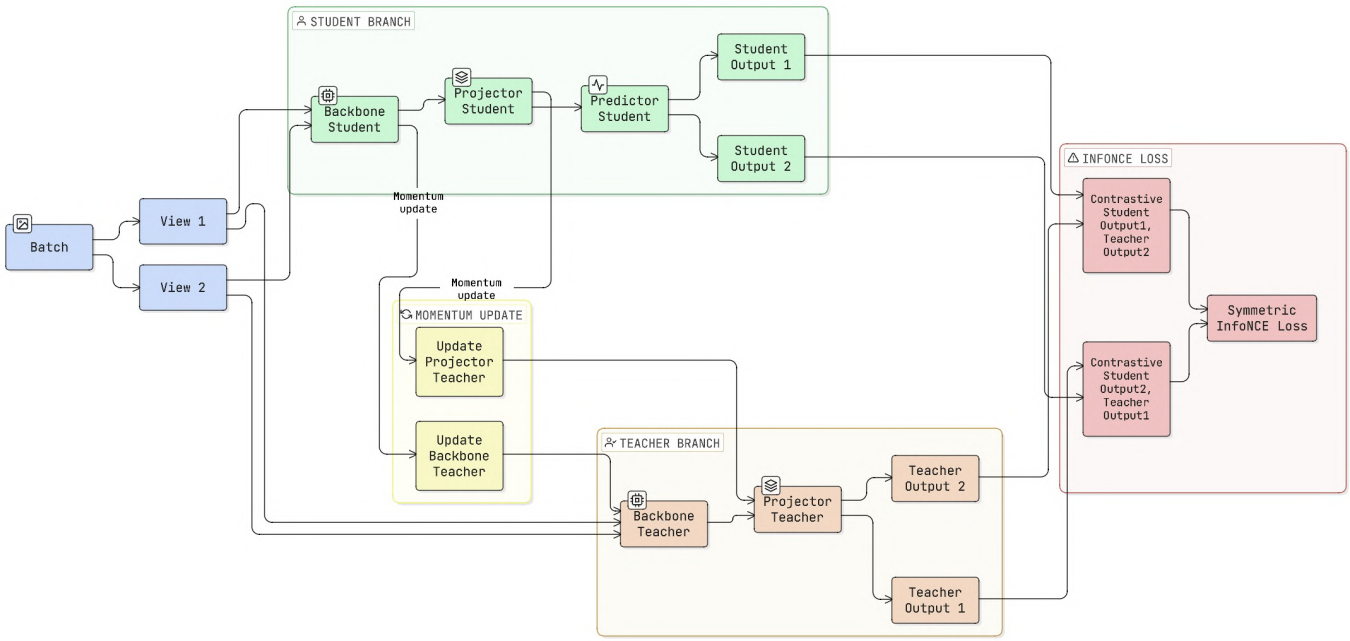

Fig. 9. Schematic overview of the MoCo v3 architecture.

D. DINO v3 (Self-Distillation with Multi-Crop)

Principle. DINO v3 casts self-supervised learning as a form of self-distillation without labels.

As illustrated in Figure 10, it employs a multi-crop strategy and a dual-branch architecture.

From a given **Batch**, it generates **Global Views** (large crops) and **Local Views** (small crops). The **Teacher Branch** – comprising a momentum-updated **tea (Frozen ViT)** backbone and two teacher heads (**DINO head** for `[CLS]` and **iBOT head** for patch tokens, both updated via EMA and not by gradients) – processes exclusively the Global Views. The **Student Branch** – comprising the **stu (ViT Backbone)** and the corresponding heads – processes the Global Views (pixel-masked for the patch objective) and all Local Views. The student's weights are actively updated via backpropagation, while the teacher backbone and heads are slowly updated via an **ema update** (Exponential Moving Average) of the student's parameters.

This design encourages the student to learn scale-invariant and context-rich representations by predicting the global context from local patches.

a) Objective (high level).: The student generates raw **Student Logits**, which are converted into probability distributions inside the loss (softmax/log-softmax with temperature). Concurrently, the teacher generates corresponding targets. In our implementation, DINO v3 optimizes a composite **Total Loss** combining four objectives:

$$\mathcal{L}_{\text{Total}} = \lambda_{[\text{CLS}]}\,\mathcal{L}_{[\text{CLS}]} + \lambda_{[\text{Patch}]}\,\mathcal{L}_{[\text{Patch}]} + \lambda_{[\text{Gram}]}\,\mathcal{L}_{[\text{Gram}]} + \lambda_{[\text{KoLeo}]}\,\mathcal{L}_{[\text{KoLeo}]} \tag{4}$$

where $\mathcal{L}_{[\text{CLS}]}$ is the distillation loss on the `[CLS]` outputs (teacher targets obtained via Sinkhorn-Knopp normalization at temperature); $\mathcal{L}_{[\text{Patch}]}$ is an iBOT-style masked patch distillation loss on patch tokens (computed only on masked regions of the global crops); $\mathcal{L}_{[\text{Gram}]}$ is the Gram anchoring loss applied to patch tokens to stabilize dense representations over long schedules; and $\mathcal{L}_{[\text{KoLeo}]}$ is a feature regularization loss encouraging a uniform and well-spaced distribution of global representations within the batch.

b) Collapse prevention.: Because DINO v3 does not use explicit negative samples (unlike contrastive methods), it relies on the combination of momentum teacher updates and non-trivial teacher target construction to avoid collapse. In our implementation, teacher logits are converted into targets via **Sinkhorn-Knopp** normalization (balanced assignments over prototypes) with temperature sharpening, and the **KoLeo** regularizer further discourages degenerate feature distributions.

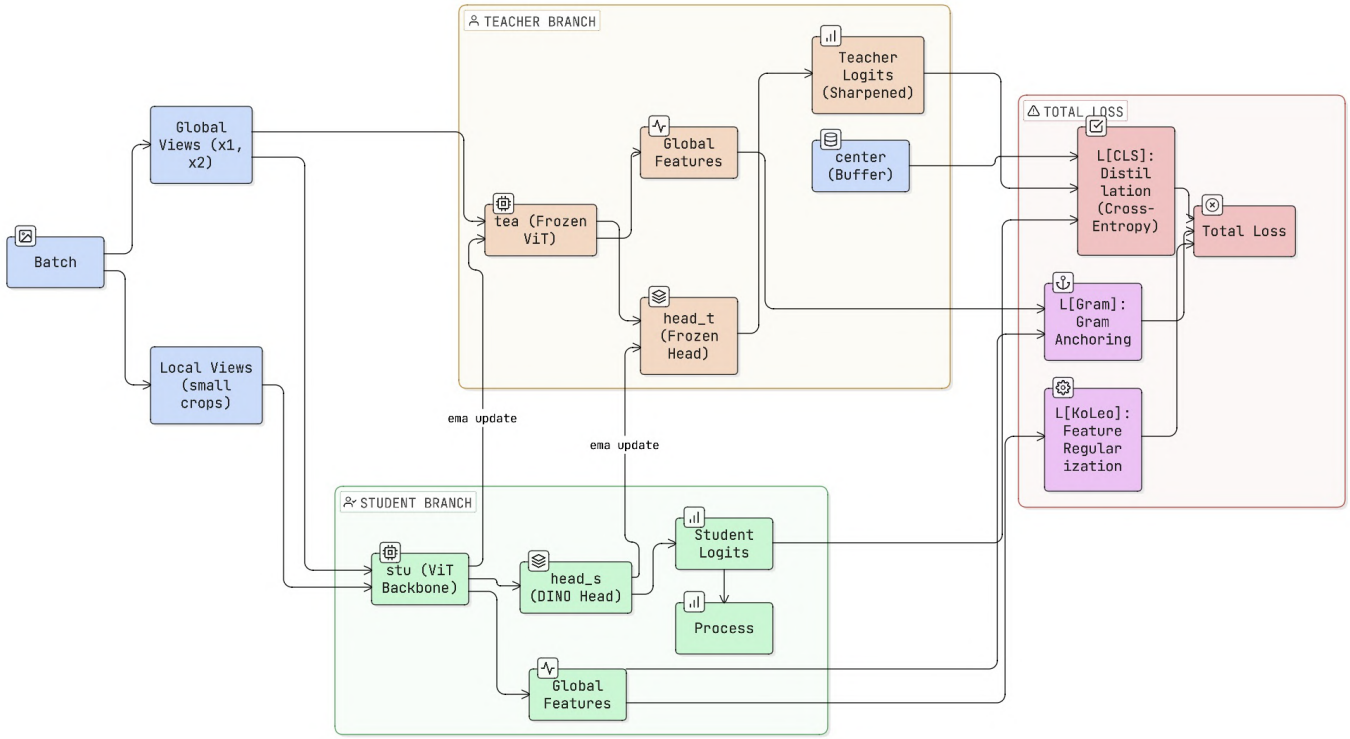

Fig. 10. Schematic overview of the DINO v3 architecture.

E. iBOT (Image BERT Pre-training with Online Tokenizer)

Principle. iBOT casts Masked Image Modeling (MIM) as a form of self-distillation. As illustrated in Figure 11, from a given **Batch** of images, it generates two augmented views (**View 1 & View 2**). The **Teacher Branch** – comprising a **Teacher Backbone** and a **Teacher Head** – processes these views intact, acting as an online tokenizer; teacher weights are not updated by gradients but via an EMA of the corresponding student weights. Concurrently, a **random patch mask** operation is applied in image space to obscure a subset of patch locations, generating **Masked View 1 & 2**. These corrupted views are fed exclusively to the **Student Branch** (**Student Backbone** and **Student Head (MLP Projection)**). Instead of reconstructing raw pixels, the student must reconstruct the semantic token distributions produced by the teacher for the masked patches.

a) Objective (high level).: The student generates both global and local representations (**Student CLS / Patch from View 1/2**), while the teacher provides the corresponding target distributions (**Teacher CLS / Patch from View 1/2**). To prevent mode collapse, the teacher's outputs are stabilized using **Center CLS & Center Patch** operations. The optimization is driven by a **Total Loss Value** that combines two distinct objectives:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{CLS}} + \mathcal{L}_{\text{Patch}} \tag{5}$$

where $\mathcal{L}_{\text{CLS}}$ represents the **Loss CLS (Cross-View Distillation)** to align the global class tokens, and $\mathcal{L}_{\text{Patch}}$ represents the **Loss Patch (Masked Patch Distillation)**, which forces the student to predict the teacher's patch-level outputs specifically for the masked regions.

b) Practical implication.: This dual distillation objective can be highly beneficial in domains like histopathology by encouraging localized morphological sensitivity (e.g., distinguishing fine-grained nuclei/cytoplasm patterns). However, the masking mechanism may require careful tuning of the mask ratio and data augmentation strength to avoid overfitting to superficial visual artifacts.

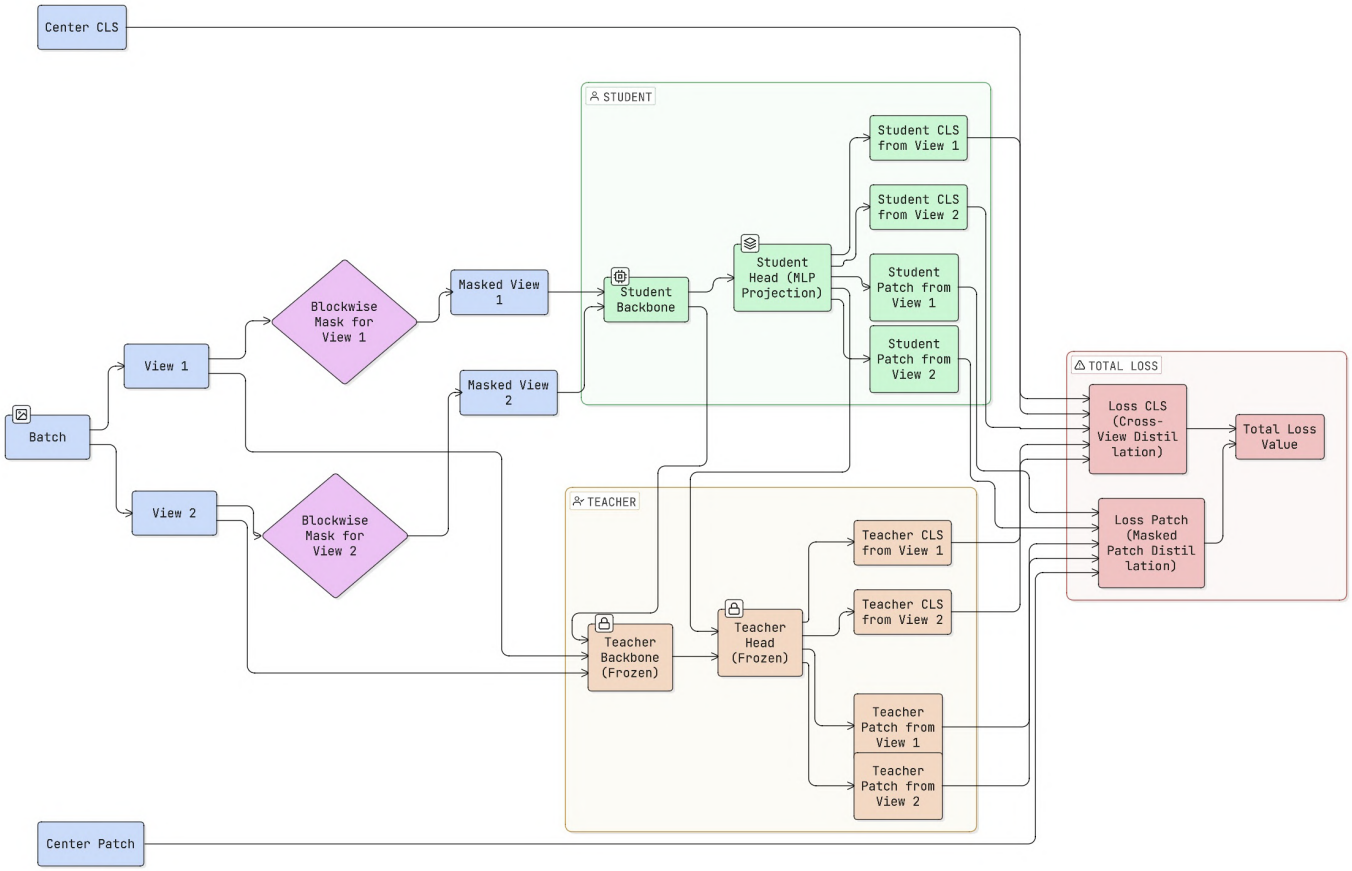

Fig. 11. Schematic overview of the iBOT architecture.

F. I-JEPA (Image-based Joint-Embedding Predictive Architecture)

Principle. I-JEPA aims to learn semantically structured representations by predicting the embeddings of missing regions from visible context, entirely in the latent space without reconstructing pixels. As illustrated in Figure 12, an **Input Image** is processed by a **Mask Generator** to define a **Context Block** and several **Target Blocks**. The **Student ViT Backbone** processes exclusively the visible context via a **Masked Forward Student** pass to extract **Context Tokens**. Concurrently, the **Teacher ViT Frozen** (whose weights are updated via an EMA of the student) processes the entire image (**Full Forward Teacher**) to produce **Full Teacher Tokens**, from which the exact **Target Tokens (Gather)** are extracted.

a) Prediction Stage and Objective.: Instead of direct contrastive alignment or pixel reconstruction, I-JEPA relies on an **I-JEPA Predictor Transformer**. This module takes the **Context Tokens** and the spatial **Predictor Position Embedding** of the target regions to generate **Pred Tokens**. The final **Loss** is an **MSE Calculation** measuring the ℓ_2 distance between the predicted tokens ($\hat{z}_t$) and the frozen target tokens (z_t):

$$\mathcal{L}_{\text{Loss}} = \|\text{stopgrad}(z_t) - \hat{z}_t\|_2^2 \,. \tag{6}$$

b) Inductive bias.: By working strictly in the latent space and discarding pixel-level generative steps, I-JEPA de-emphasizes precise low-level details and background information. In histopathology, this can reduce the pressure to encode fine textures or staining artifacts and may encourage stronger focus on higher-level morphology (e.g., overall cellular structures and spatial relationships). However, maximizing its potential requires a carefully designed context/target masking strategy to avoid losing highly localized discriminative cues.

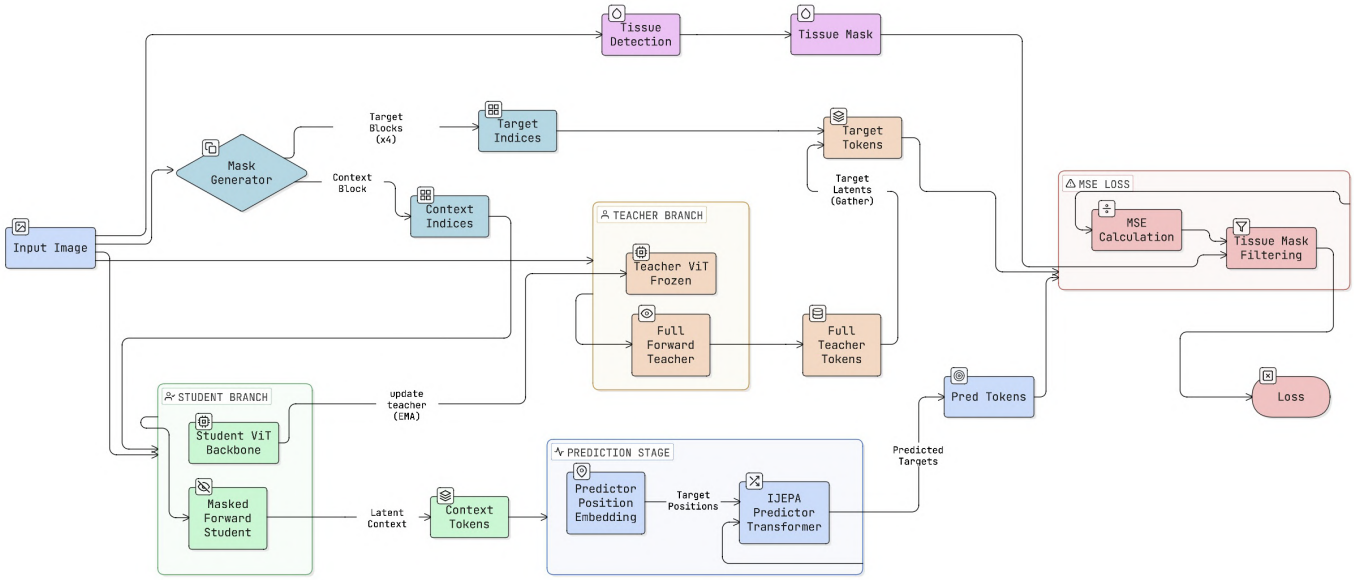

Fig. 12. Schematic overview of the I-JEPA architecture.

APPENDIX B
ABLATION STUDY DETAILS

A. MoCo v3

TABLE VI
COMMON CONFIGURATION FOR MOCO V3 (CORE SETTINGS).

Category	Setting	Value
Data & Optimization		
	Input image size	224×224
	Batch size (SSL)	64
	Optimizer	AdamW
	AdamW β_1, β_2	$(0.9, 0.999)$
	Weight decay	0.05
	Learning rate schedule	Cosine Decay
	Mixed precision (AMP)	Yes
Backbone		
	Architecture	ViT-S/16
	Patch size	16×16
SSL Specifics		
	Hidden dimension	4096
	Projection dimension	256
	Sync BatchNorm	No
	Multi-crop	No
	Momentum (m) schedule	Increase to 1.0

TABLE VII
MOCO V3 ABLATIONS. ONLY HYPERPARAMETERS VARYING ACROSS EXPERIMENTS ARE LISTED.

ID	Gaussian Blur Prob.	Grayscale Prob.	Color Jitter Strength	Temp. τ	Teacher Temp. Schedule	LR	Grad. Accum.	EMA m
abl01	0.1	0.2	0.4	0.2	None	3e-4	16	0.996
abl02	0.1	0.2	0.4	0.1	None	3e-4	16	0.996
abl03	0.1	0.2	0.4	0.4	None	3e-4	16	0.996
abl04	0.1	0.2	0.4	0.2	None	3e-4	16	0.99
abl05	0.1	0.2	0.4	0.2	None	3e-4	16	0.999
abl06	0.1	0.2	0.4	0.2	$0.04 \rightarrow 0.2$	3e-4	16	0.996
abl07	0.1	0.2	0.4	0.2	$0.2 \rightarrow 0.2$	3e-4	16	0.996
abl08	0.2	0.2	0.6	0.2	None	3e-4	16	0.996
abl09	0.1	0.1	0.2	0.2	None	3e-4	16	0.996
abl10	0.1	0.2	0.4	0.2	None	1.5e-4	8	0.996

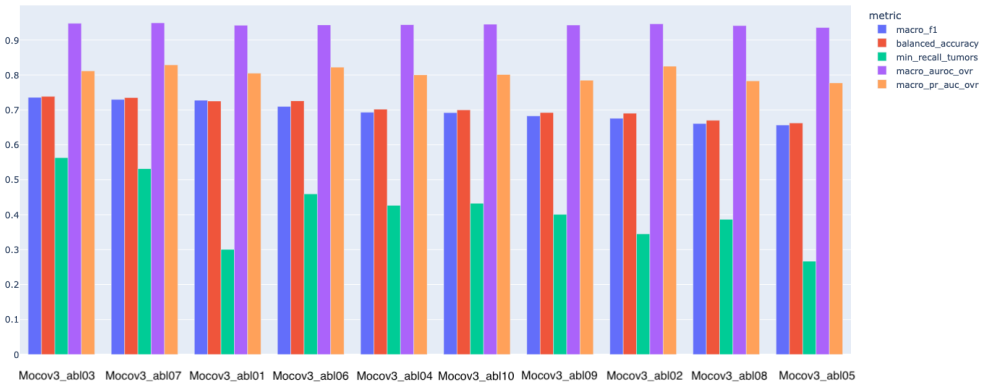

Fig. 13. **MoCo v3 Hyperparameter Sensitivity.** Downstream performance metrics for ablations on augmentation intensity, temperature, and momentum.

B. DINO v3

TABLE VIII
COMMON CONFIGURATION FOR DINO V3 (CORE SETTINGS).

Category	Setting	Value
Multi-crop Strategy		
	Global crop size	224^2
	Local crop size	96^2
	Global scale range	$[0.14, 1.0]$
Augmentation		
	Gaussian blur prob.	0.1 (global), 0.3 (local)
	Solarization prob.	0.0 (global), 0.2 (local)
	Color jitter strength	0.4
Optimization		
	Batch size (SSL)	64
	Max teacher updates	3 (GRAM)
	EMA momentum	Sched. to 1.0
	Bottleneck dim	256

TABLE IX
DINO V3 ABLATIONS. ONLY VARYING HYPERPARAMETERS ARE LISTED.

ID	Local Scale	Number Local	GRAM Start	GRAM Update	Prototypes	GRAM λ	Loss λ (iBOT)	Loss λ (KoLeo)	Loss λ (DINO v3)
abl01	$[0.05, 0.12]$	6	0.7	10k iters	8192	1.0	1.0	0.1	1.0
abl02	$[0.05, 0.12]$	6	0.7	10k iters	8192	0.0	0.0	0.0	1.0
abl03	$[0.04, 0.10]$	8	0.5	10k iters	8192	1.5	1.4	0.1	0.6
abl04	$[0.05, 0.12]$	6	0.7	10k iters	8192	0.0	1.0	0.1	1.0
abl05	$[0.05, 0.12]$	6	0.5	5k iters	8192	2.0	1.0	0.1	1.0
abl06	$[0.05, 0.12]$	6	0.7	10k iters	8192	1.0	0.0	0.1	1.0
abl07	$[0.05, 0.12]$	6	0.7	10k iters	8192	1.0	1.5	0.1	0.7
abl08	$[0.05, 0.12]$	6	0.7	10k iters	4096	1.0	1.0	0.05	1.0

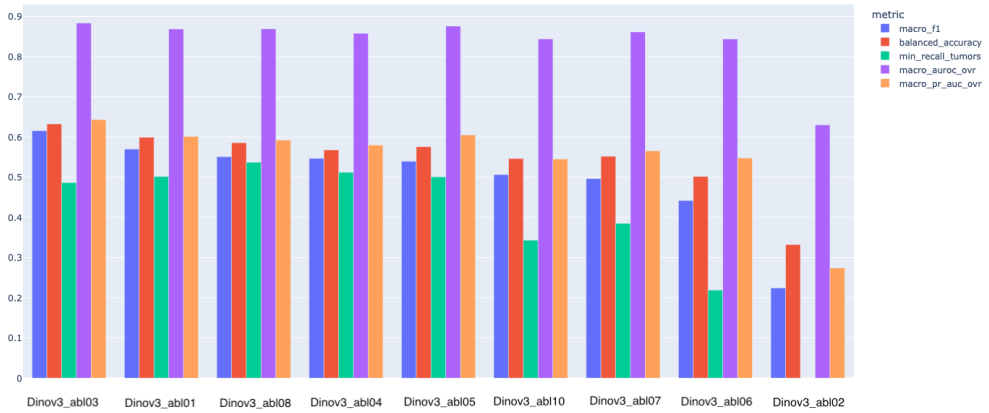

Fig. 14. **DINO v3 Component Analysis.** Comparison of classification metrics across variations in multi-crop strategy (scale/number) and loss term weighting.

C. iBOT

TABLE X
COMMON CONFIGURATION FOR IBOT (CORE SETTINGS).

Category	Setting	Value
Self-Supervised Learning		
	Hidden dimension	4096
	Centering momentum	0.9
	Student Temp. τ_s	0.2
	Student Patch Temp. $\tau_{s,p}$	0.2
	EMA momentum	Sched. to 1.0
Augmentation		
	Blur / Gray / Jitter	0.1 / 0.2 / 0.4

TABLE XI
IBOT ABLATIONS. VARYING HYPERPARAMETERS.

ID	Prototypes	Mask Ratio	Teacher Temp. τ_t	Teacher Patch Temp. $\tau_{t,p}$	Base EMA m	Final EMA m
abl01	8192	0.3	0.07	0.07	0.996	1.0
abl02	4096	0.3	0.07	0.07	0.996	1.0
abl03	16384	0.3	0.07	0.07	0.996	1.0
abl04	8192	0.5	0.07	0.07	0.996	1.0
abl05	8192	0.2	0.07	0.07	0.996	1.0
abl06	8192	0.3	0.04	0.04	0.996	1.0
abl07	8192	0.3	0.10	0.10	0.996	1.0
abl08	8192	0.3	0.07	0.07	0.990	0.999

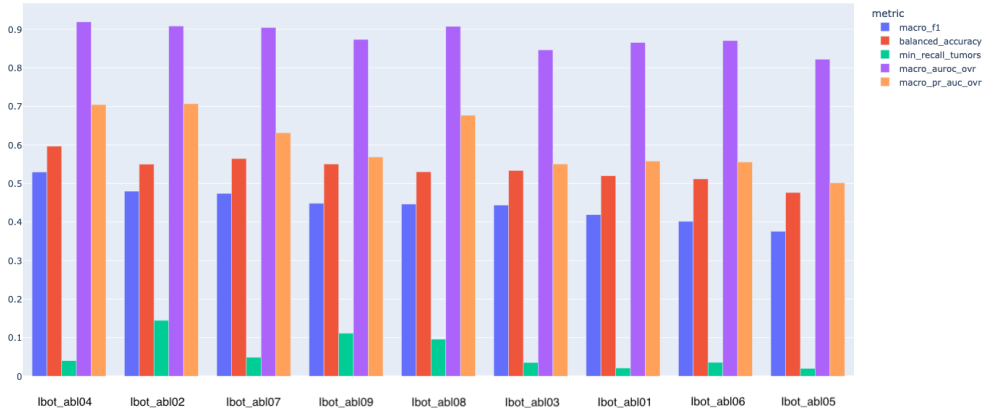

Fig. 15. **iBOT Masking and Distillation Analysis.** Impact of masking ratio and teacher temperature schedules on downstream classification performance.

D. I-JEPA

TABLE XII
COMMON CONFIGURATION FOR I-JEPA (CORE SETTINGS).

Category	Setting	Value
Architecture		
	Backbone	ViT-S/16
	Projection Head Dim	256
	Hidden Dim	4096
Training		
	Optimizer	AdamW
	Betas	$(0.9, 0.999)$
	Mixed Precision	Yes
	Background Std. Thresh.	0.02

TABLE XIII
I-JEPA ABLATIONS: TRAINING AND AUGMENTATION HYPERPARAMETERS.

ID	Gaussian Blur Prob.	Grayscale Prob.	Color Jitter Strength	EMA m Schedule	Base EMA	LR	W. Decay
abl03	0.1	0.2	0.4	Yes	0.996	3e-4	0.05
abl04	0.1	0.2	0.4	Yes	0.996	3e-4	0.05
abl05	0.1	0.2	0.4	Yes	0.996	3e-4	0.05
abl06	0.1	0.2	0.4	Yes	0.996	3e-4	0.05
abl07	0.1	0.2	0.4	Yes	0.996	3e-4	0.05
abl08	0.1	0.2	0.4	No	0.990	3e-4	0.05
abl09	0.1	0.2	0.4	Yes	0.998	3e-4	0.05
abl10	0.1	0.2	0.4	Yes	0.996	5e-4	0.10
abl11	0.0	0.0	0.0	Yes	0.996	3e-4	0.05

TABLE XIV
I-JEPA ABLATIONS: MASKING AND PREDICTOR ARCHITECTURE.

ID	Ctx Scale	Tgt Scale	Target Blocks	Pred. Depth	Pred. Dim	Pred. Heads
abl03	Default	Default	4	12	384	6
abl04	Default	Default	6	6	384	6
abl05	Default	Default	2	6	384	6
abl06	Default	Default	4	6	512	8
abl07	Default	Default	4	6	256	4
abl12	$[0.75, 1.0]$	$[0.15, 0.25]$	4	6	384	6

Note: "Default" scales refer to context $[0.85, 1.0]$ and target $[0.15, 0.2]$.

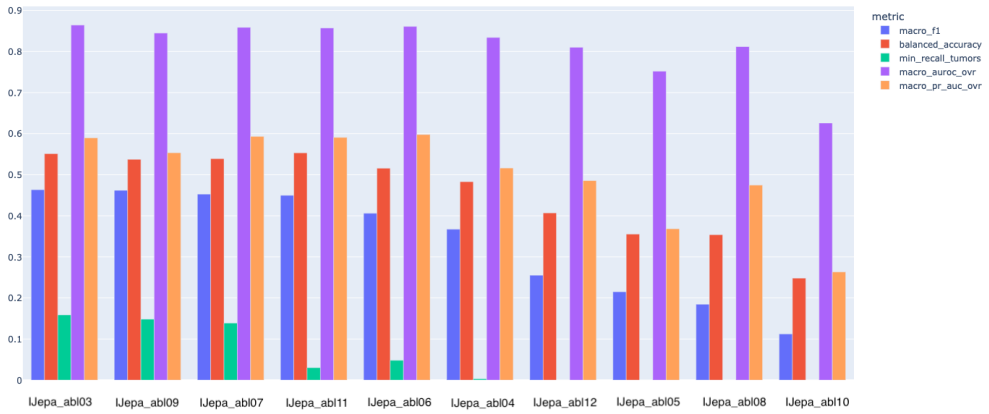

Fig. 16. **I-JEPA Architecture and Masking Ablations.** Performance metrics evaluating predictor depth/width and context/target masking scales.

APPENDIX C
CONFUSION MATRICES

This appendix reports additional confusion matrices for all evaluated model families. Patch-level evaluation uses five classes (ccRCC, pRCC, CHROMO, ONCO, NOT_TUMOR). Patient-level evaluation uses tumor-only aggregation (NOT_TUMOR excluded).

A. Patch-level confusion matrices (test split)

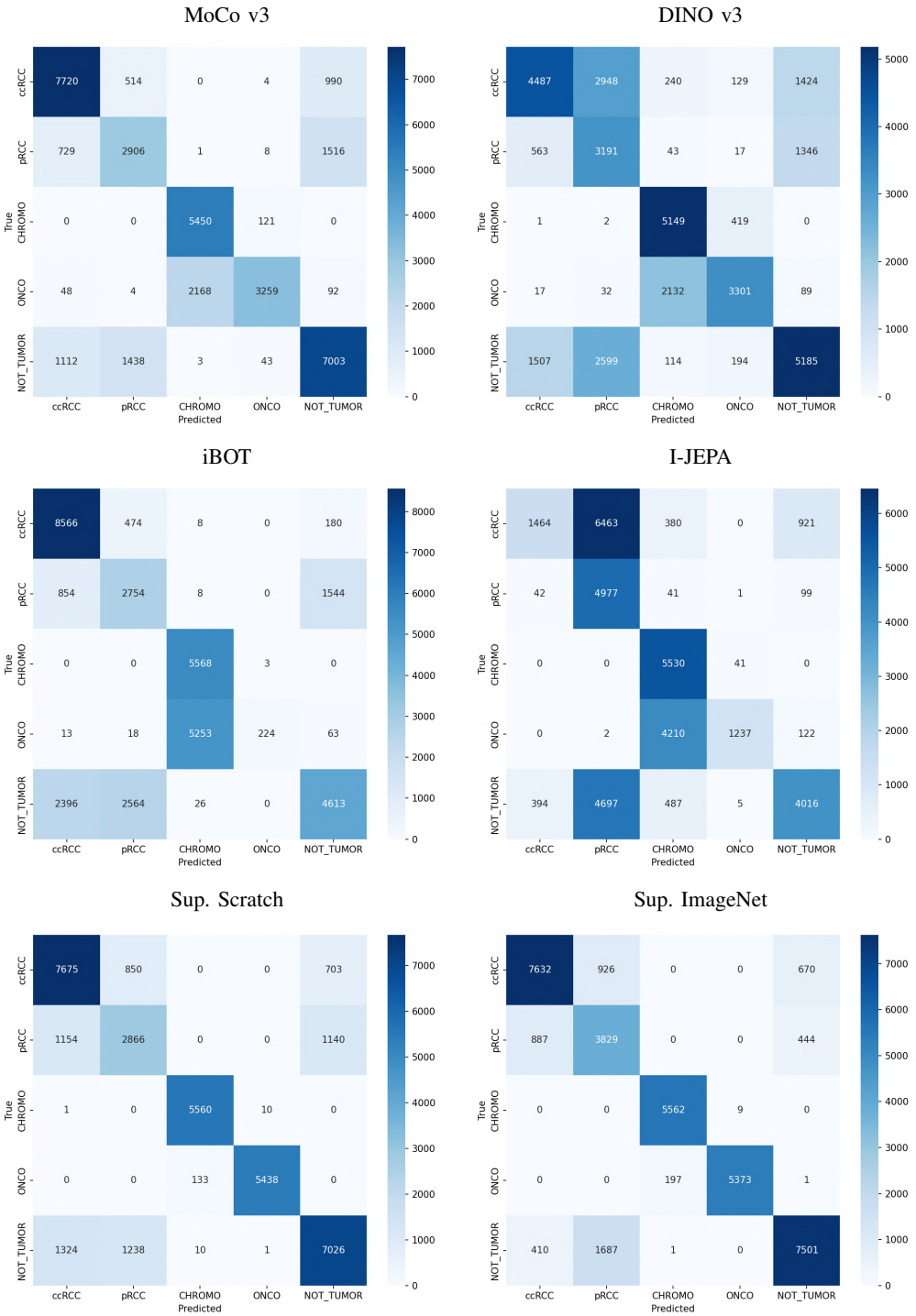

Fig. 17. Patch-level confusion matrices for all evaluated models (5-way test task).

B. Patient-level confusion matrices (tumor-only aggregation)

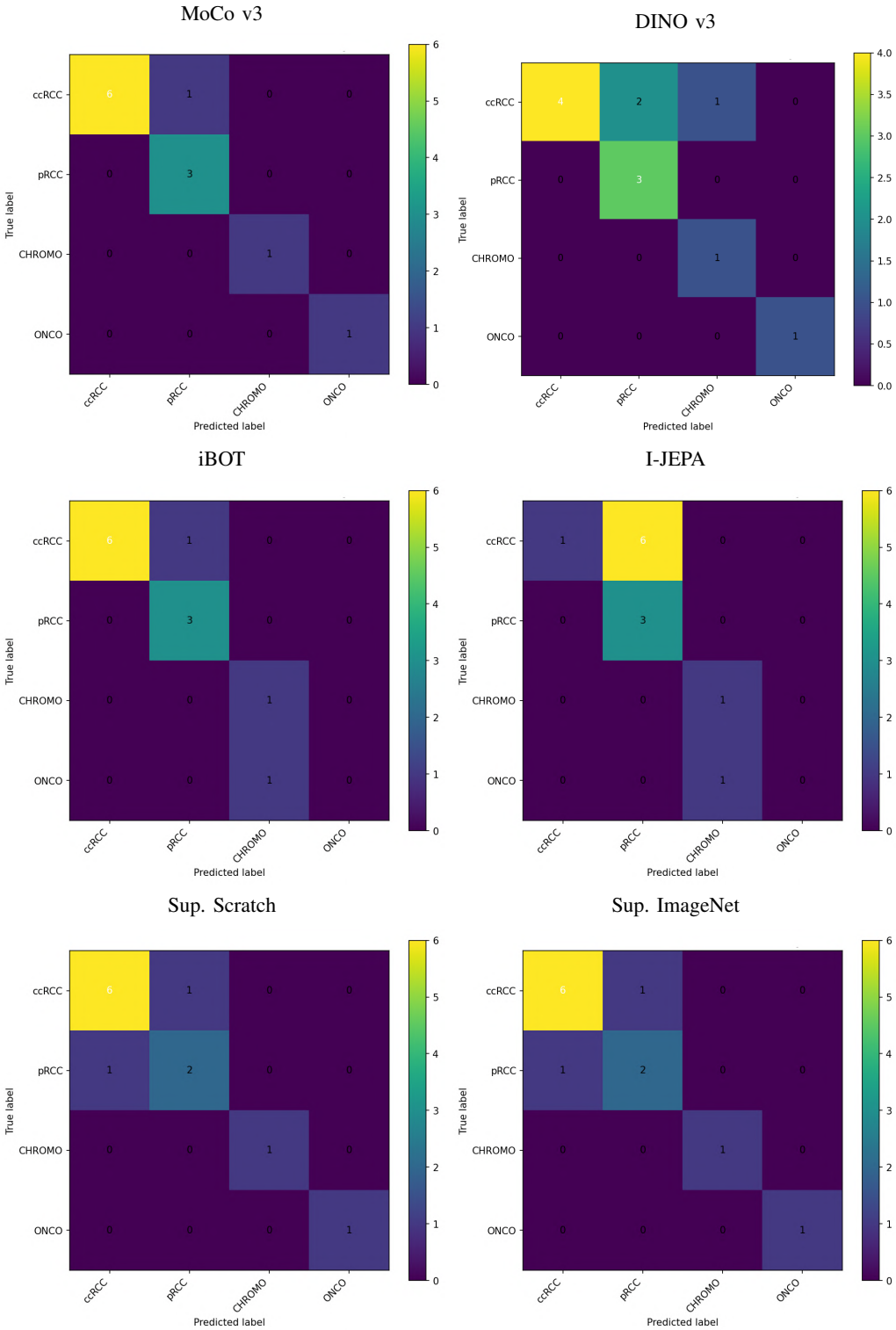

Fig. 18. Patient-level confusion matrices for all evaluated models (tumor-only aggregation).

APPENDIX D
CONCEPT BANK

A. PLIP Concepts by Class (Primary + Confounds)

TABLE XV: Top PLIP concepts per class from the shortlist (ranked by AUC OvR). Primary concepts are class-consistent; confounds are high-ranked cross-class concepts. Descriptions are shortened from concept prompts.

ID	Concept	Description
ccRCC		
Primary		
C001	solid_clear	Solid clear sheets; few capillaries.
C002	macrocystic_change	Large cystic/pseudocystic spaces lined by clear cells; thin septa.
C003	clear_diffuse	Optically empty cytoplasm with small round nuclei.
C004	clear_capillaries	Fine "chicken-wire" capillary network between clear-cell nests.
C005	tubular_acinar	Tubules/acini of clear cells with small lumina and delicate vessels.
C006	alveolar_nested	Rounded nests separated by capillary-rich septa (alveolar/nested).
C007	microcystic	Small microcysts/vacuoles between clear tumor cells.
Confounds		
C008	solid_sheets_pale	Broad sheets of pale eosinophilic polygonal cells; distinct borders.
C009	dediff_rhabdoid	Large rhabdoid cells with eccentric nuclei and dense eosinophilic inclusions.
C010	neutrophils_acute	Abundant neutrophils/acute exudate in stroma or lumina.
C011	grade_low_isup12	Small uniform nuclei with inconspicuous nucleoli (grading cue).
C012	archipelagenous	Round oncocytic islands separated by pale edematous/myxoid stroma.
pRCC		
Primary		
C013	papillary_fronds	True papillae with fibrovascular cores lined by neoplastic epithelium.
C014	type1	Small cuboidal cells, scant pale cytoplasm, low-grade nuclei on slender papillae.
C015	pseudostratified	Crowded lining with nuclear pseudostratification along papillary surfaces.
C016	hemosiderin_macrophages	Coarse golden-brown hemosiderin pigment in macrophages within papillary cores.
C017	hyalinized_cores	Thick glassy hyalinized fibrovascular cores beneath epithelium.
C018	tubulopapillary	Short papillae fused with tubular structures (tubulo-papillary).
C019	foamy_macrophages	Numerous foamy macrophages with vacuolated cytoplasm in cores/lumina.
C020	hobnail_nuclei	Nuclei protrude into lumen producing hobnail luminal contour.
Confounds		
C021	peritumoral_kidney	Interface with preserved tubules/glomeruli adjacent to tumor.
C022	medulla_collecting_vasa_recta	Parallel collecting ducts and thin vasa recta; medullary pattern.
C023	hemorrhage	Extensive fresh/organizing hemorrhage; non-specific.
C024	lymphocytes	Dense lymphocytes in stroma/interface; non-specific.
C025	thick_vessels	Thick-walled hyalinized vessels between oncocytic nests.
chRCC (CHROMO)		
Primary		
C026	pale_reticulated	Pale finely reticulated eosinophilic cytoplasm with subtle vesicular texture.
C027	oncocytic_variant	More eosinophilic cytoplasm but retains halos and/or plant-cell borders.
C028	perinuclear_halos	Conspicuous perinuclear clearing with crisp cytoplasmic borders.

ID	Concept	Description
C029	microvesicular_cytoplasm	Pale microvesicular/reticulated cytoplasm giving finely vacuolated look.
C030	binucleation	Frequent binucleation with raisinoid nuclear irregularity and sharp membranes.
Confounds		
C031	glomerulus_bowman	Glomerular tuft with Bowman space; surrounding tubules; no tumor nests.
C032	psammoma_bodies	Small round concentrically laminated calcifications in papillae/stroma.
C033	mixed_cytoplasm	Mostly clear cells with a substantial eosinophilic granular subset in same nests.
C034	grade_high_isup34	Enlarged pleomorphic nuclei with prominent nucleoli (grading cue).
C036	round_nuclei_no_halos	Oncocytic nests with round nuclei; absent perinuclear halos.
Oncocytoma (ONCO)		
Primary		
C036	round_nuclei_no_halos	Oncocytic nests with round nuclei and absent perinuclear halos.
C037	bland_round_nuclei	Uniform round nuclei, minimal pleomorphism; lacks raisinoid wrinkling/strong halos.
C038	nested_trabecular	Nests and trabeculae of oncocytic cells separated by delicate stroma.
C039	uncommon_binucleation_no_necrosis	Binucleation uncommon; no zonal coagulative necrosis; bland cytology.
C040	oncocytic_cytoplasm	Dense granular intensely eosinophilic oncocytic cytoplasm with relatively uniform nuclei.
C041	polygonal_cells_prominent_nucleoli	Large polygonal oncocytes with conspicuous nucleoli and dense granular cytoplasm.
C042	tubulocystic	Small tubules/cysts lined by oncocytes; granular eosinophilic cytoplasm and bland nuclei.
C043	nests_small_capillaries	Tight nests with many small capillaries in intervening stroma.
Confounds		
C020	hobnail_nuclei	Nuclei protrude into papillary lumen producing hobnail contour.
C033	mixed_cytoplasm	Mixed clear/eosinophilic cytoplasm within the same nests (cross-subtype).
C032	psammoma_bodies	Concentric laminated calcifications; often associated with papillary features.
C047	type2	Larger eosinophilic cells with higher grade and prominent nucleoli.
C034	grade_high_isup34	Enlarged pleomorphic nuclei with prominent nucleoli (grading cue).
Not-tumor (NOT_TUMOR)		
Primary		
C049	interstitial_fibrosis_atrophy	Interstitial fibrosis with atrophic tubules; preserved architecture; no malignant nests.
C024	lymphocytes	Dense lymphocytes in stroma or interface; non-specific.
C051	scar_like_stroma	Dense collagenized scar-like stroma with low cellularity; non-specific.
C052	distal_collecting	Regular ducts with paler cuboidal cells and clean lumina.
C053	hyaline_basement_membranes	Thick hyalinized basement-membrane-like material around tubules/structures.
C021	peritumoral_kidney	Interface with preserved tubules and glomeruli adjacent to tumor.
C055	fibrous_septa	Broad collagen-rich septa separating tissue elements.
C023	hemorrhage	Fresh/organizing hemorrhage; non-specific signal.
Confounds		
C017	hyalinized_cores	Thick hyalinized fibrovascular cores beneath papillary epithelium.
C058	necrosis_coagulative	Sharp transition to eosinophilic “ghost cells” and granular debris.
C059	peripheral_rim_edematous	Peripheral rim of oncocytes surrounding pale edematous stromal tissue.
C060	dediff_sarcomatoid	Malignant spindle-cell fascicles with loss of epithelial architecture.

APPENDIX E
PLIP CONCEPT AUDIT

Setup. For each class, concepts are evaluated in a one-vs-rest (OvR) setting by comparing the target class against all remaining classes. We rank concepts by $\mathrm{AUROC}_{\mathrm{ovr}}$ and apply filters ($\mathrm{AUROC}_{\mathrm{ovr}} \geq 0.6$, $\mathrm{AP}_{\mathrm{ovr}} \geq 0.0$, Cohen's $d \geq 0.3$), where $\mathrm{AUROC}_{\mathrm{ovr}}$ measures ranking-based separability, $\mathrm{AP}_{\mathrm{ovr}}$ summarizes precision–recall behavior under class imbalance, and Cohen's d measures the effect size between concept scores in the target class and the rest. These thresholds define the default filtered candidate pool: passing them is necessary to enter the filtered ranking, but not sufficient to appear in the final shortlist, because concepts must also rank highly within the target class. We then select up to $k_{\mathrm{primary}} = 8$ diagnostic concepts and up to $k_{\mathrm{confounds}} = 5$ high-rank cross-class concepts; therefore, a class may contain fewer than 13 concepts if fewer candidates remain after filtering/ranking. When the default thresholds would otherwise empty a class, the implementation falls back to a relaxed ranking so that no class is left without inspectable concepts. The bar plots below should therefore not be read as the final shortlist itself: they show the top-12 concepts per class for a given metric (AUROC or $\Delta = \mu_{\mathrm{pos}} - \mu_{\mathrm{rest}}$), so their counts and membership can differ from the semantic shortlist reported in Appendix Table XV. In practice, this keeps concepts that are not only highly ranked but also show a real score separation, while the confounds highlight plausible but non-specific morphology that may contribute to ambiguity.

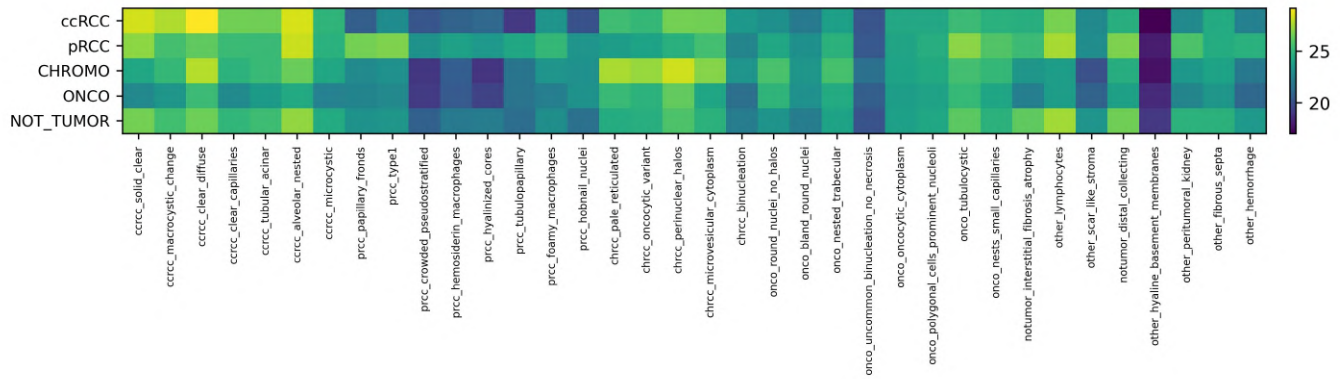

Fig. 19. Heatmap mean score class–concept (PLIP).

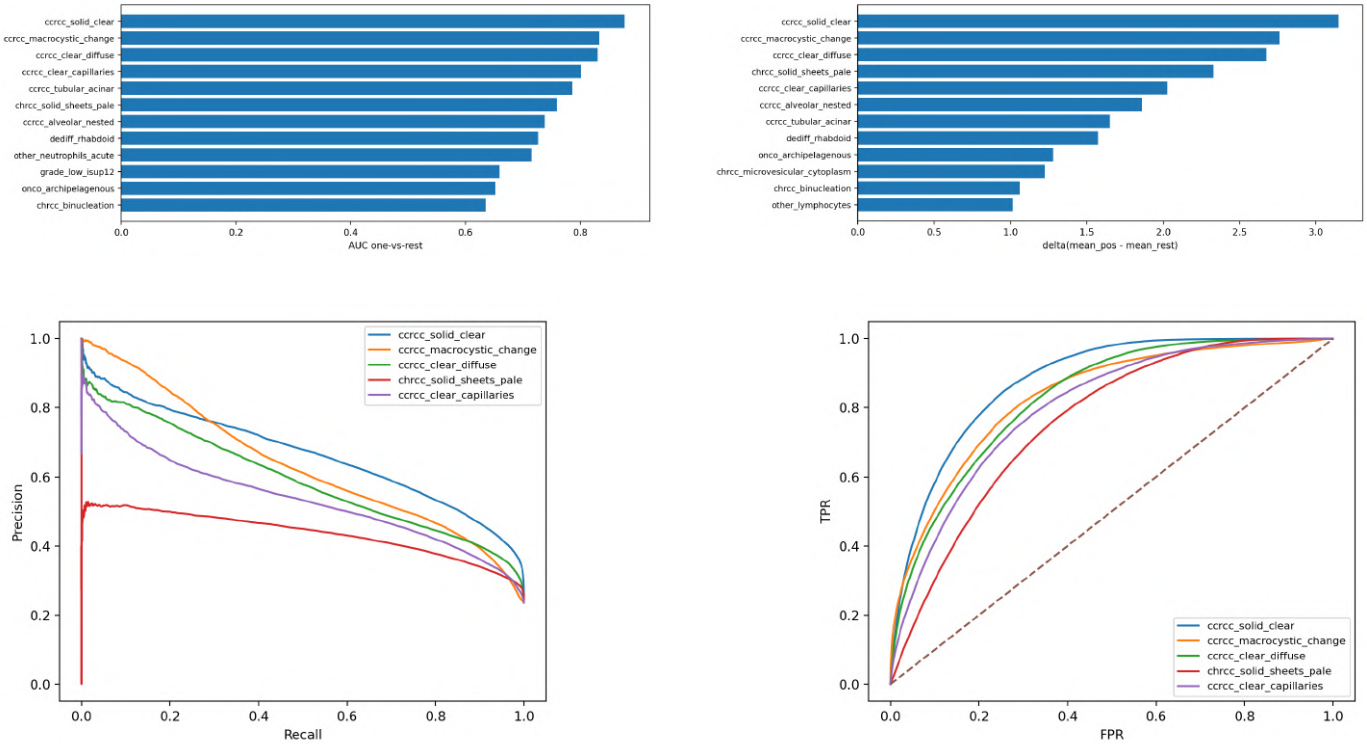

Fig. 20. PLIP concept performance for ccRCC. The two bar plots report the top-12 concepts for this class by $\mathrm{AUROC}_{\mathrm{ovr}}$ and by $\Delta = \mu_{\mathrm{pos}} - \mu_{\mathrm{rest}}$, respectively; they are metric-specific visual summaries and do not necessarily coincide with the final primary/confound shortlist.

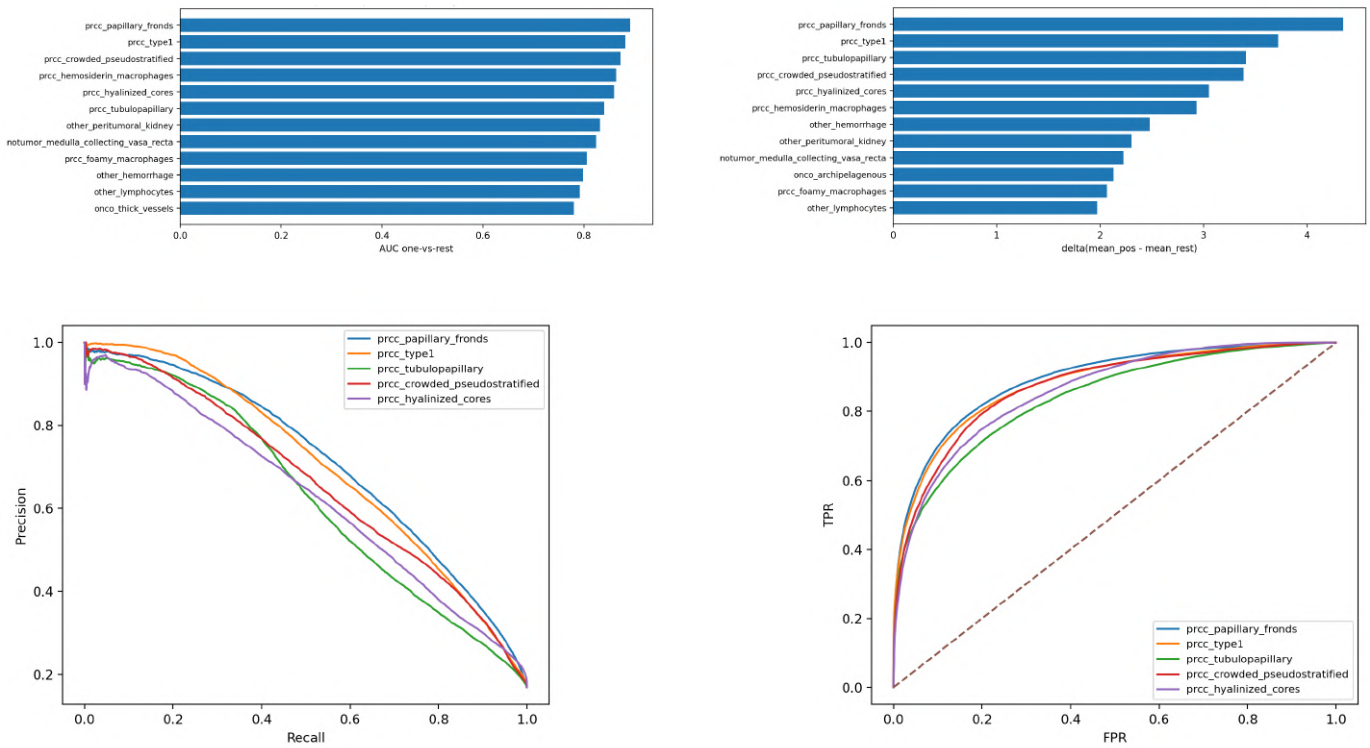

Fig. 21. PLIP concept performance for pRCC. The two bar plots report the top-12 concepts for this class by $\mathrm{AUROC}_{\mathrm{ovr}}$ and by $\Delta = \mu_{\mathrm{pos}} - \mu_{\mathrm{rest}}$, respectively; they are metric-specific visual summaries and do not necessarily coincide with the final primary/confound shortlist.

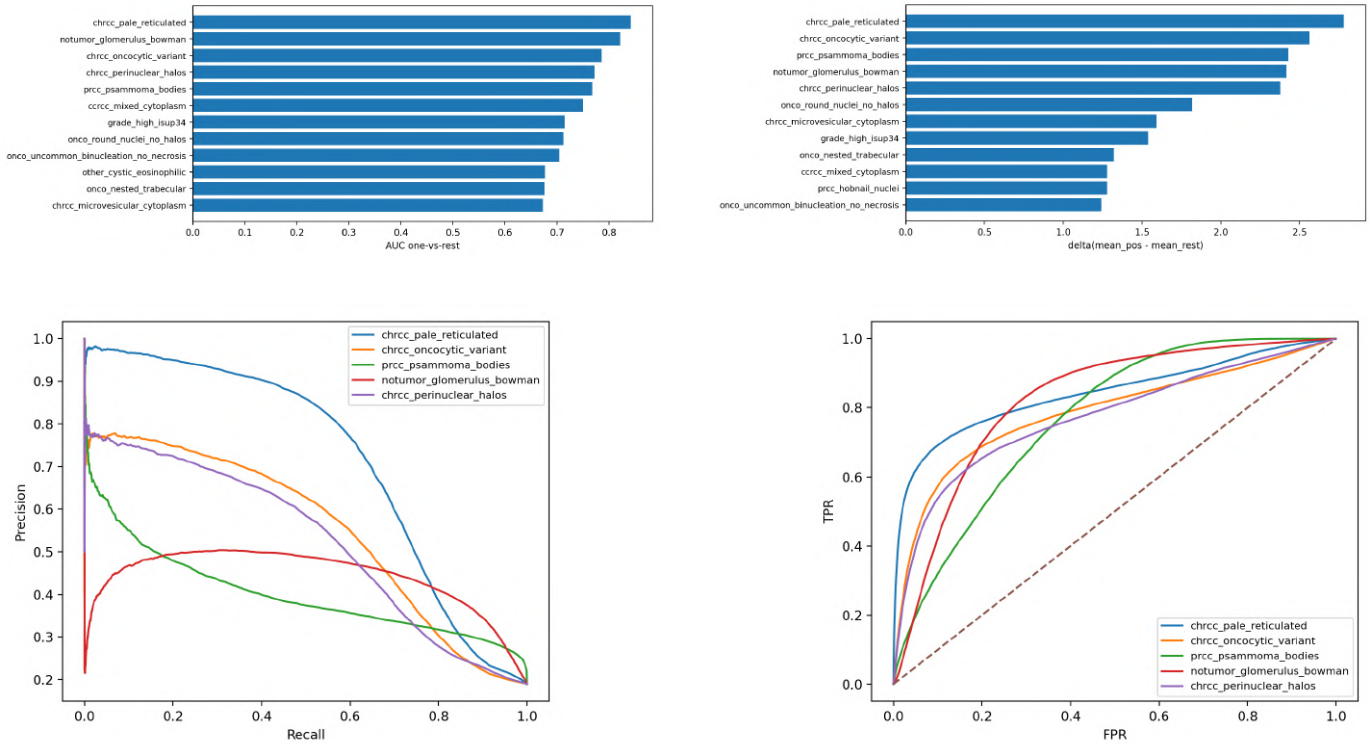

Fig. 22. PLIP concept performance for CHROMO. The two bar plots report the top-12 concepts for this class by $\mathrm{AUROC}_{\mathrm{ovr}}$ and by $\Delta = \mu_{\mathrm{pos}} - \mu_{\mathrm{rest}}$, respectively; they are metric-specific visual summaries and do not necessarily coincide with the final primary/confound shortlist.

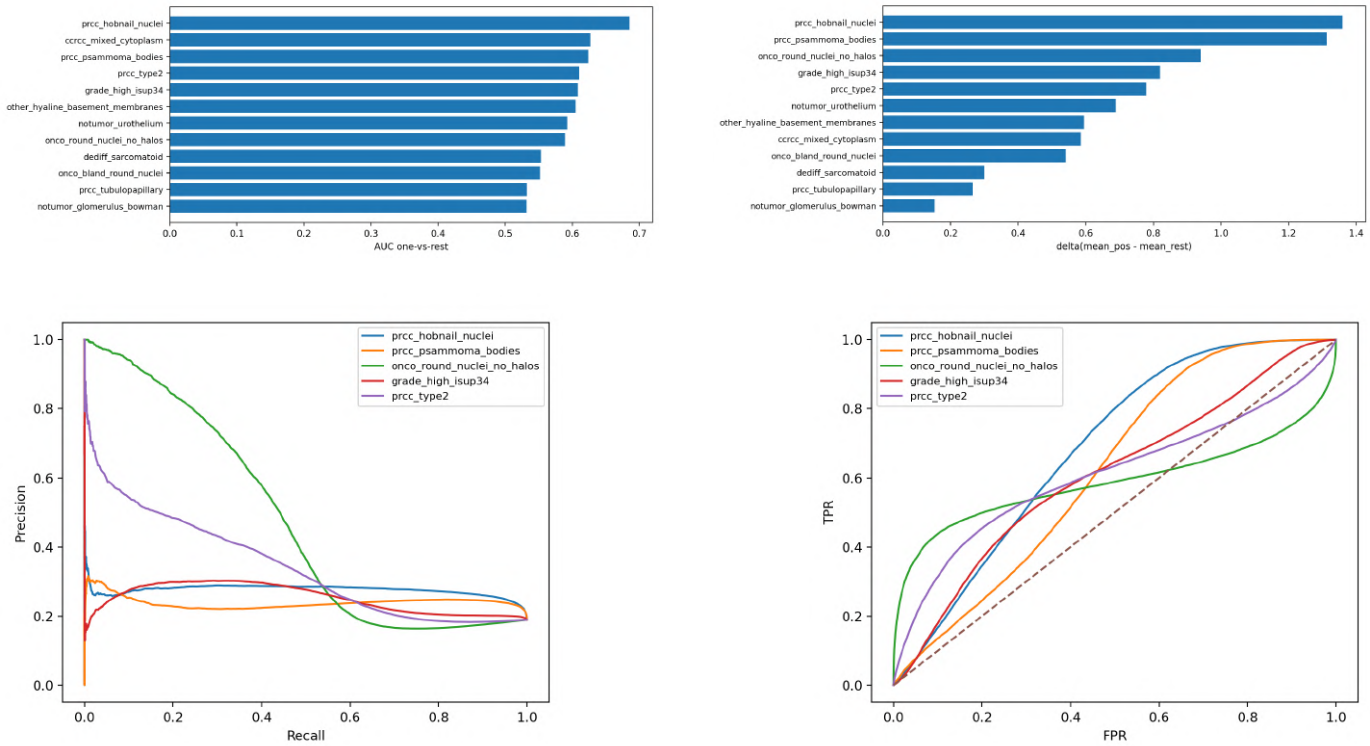

Fig. 23. PLIP concept performance for ONCO. The two bar plots report the top-12 concepts for this class by $\mathrm{AUROC}_{\mathrm{ovr}}$ and by $\Delta = \mu_{\mathrm{pos}} - \mu_{\mathrm{rest}}$, respectively; they are metric-specific visual summaries and do not necessarily coincide with the final primary/confound shortlist.

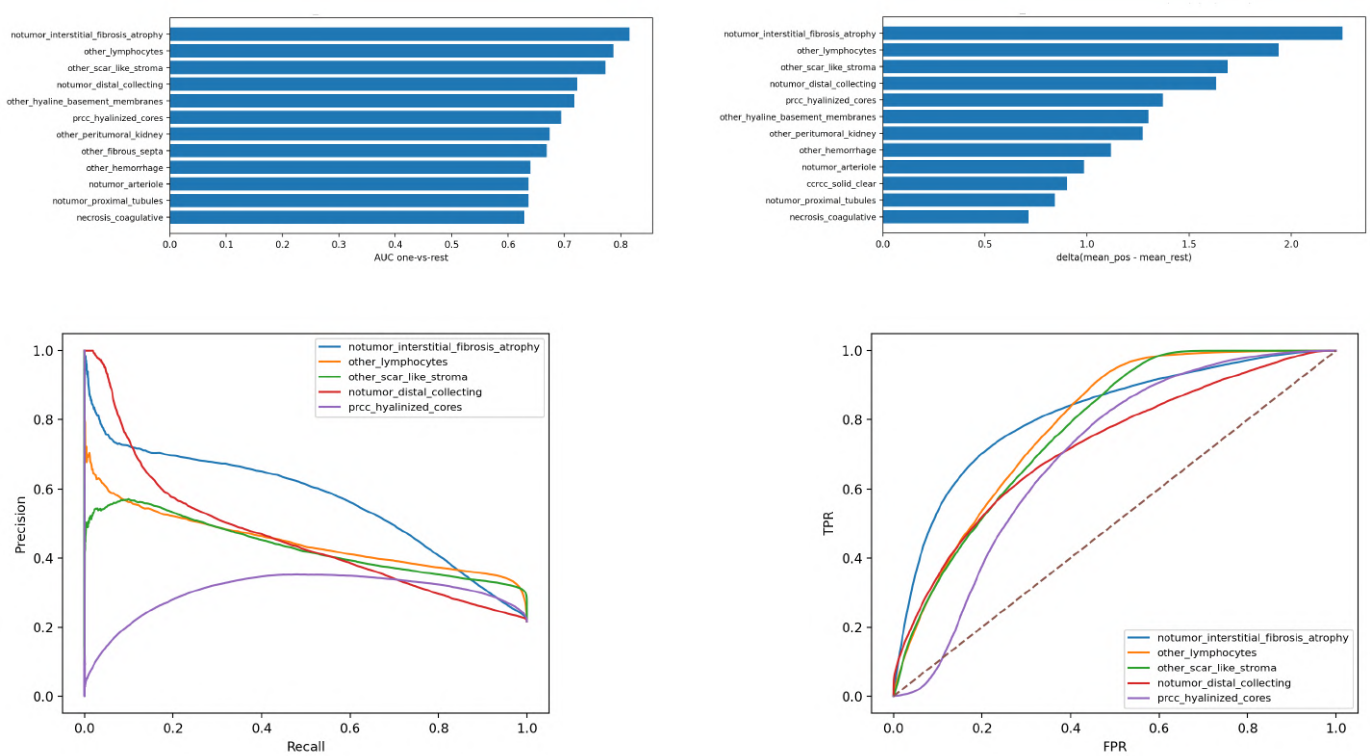

Fig. 24. PLIP concept performance for NOT_TUMOR. The two bar plots report the top-12 concepts for this class by $\mathrm{AUROC}_{\mathrm{ovr}}$ and by $\Delta = \mu_{\mathrm{pos}} - \mu_{\mathrm{rest}}$, respectively; they are metric-specific visual summaries and do not necessarily coincide with the final primary/confound shortlist.

APPENDIX F
PLIP CONCEPT SPATIAL GROUNDING

This appendix provides additional qualitative examples of concept-level spatial grounding using PLIP-based concept scoring combined with transformer attention rollout. In each case, PLIP concepts are scored on an attention-guided patch view, obtained by thresholding the rollout map (top-quantile), localizing salient connected components, converting them into ROI bounding-box contours, and overlaying these contours on the original patch (the resulting patch + ROI contours is passed to PLIP). Captions report the ground-truth class (T) and predicted class (P). Reported concept scores are PLIP logits, i.e., scaled cosine similarities between normalized PLIP image and text embeddings using the model's learned logit scale; they are not probabilities (Appendix D Table XV).

A. MoCo v3

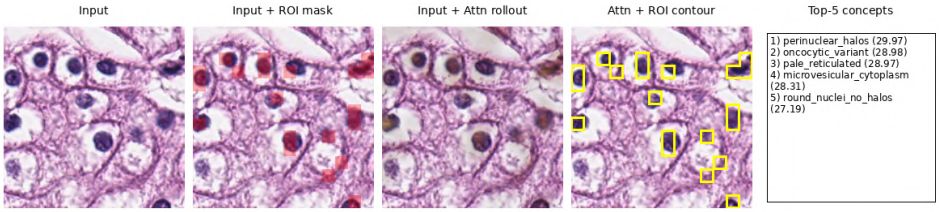

Fig. 25. Extended PLIP concept grounding example for MoCo v3 (T=CHROMO, P=CHROMO). Top concepts in panel (e): perinuclear halos (C028), oncocytic variant (C027), pale reticulated (C026), microvesicular cytoplasm (C029), round nuclei no halos (C036). PLIP concepts are class-consistent with the metadata label.

B. DINO v3

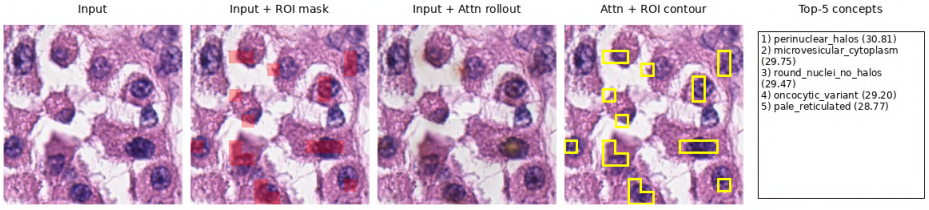

Fig. 26. PLIP concept grounding example for DINO v3 (T=ONCO, P=CHROMO). Top concepts in panel (e): perinuclear halos (C028), microvesicular cytoplasm (C029), round nuclei no halos (C036), oncocytic variant (C027), pale reticulated (C026). The top concepts emphasize oncocytic/chromophobe morphology, consistent with ONCO–CHROMO ambiguity.

C. iBOT

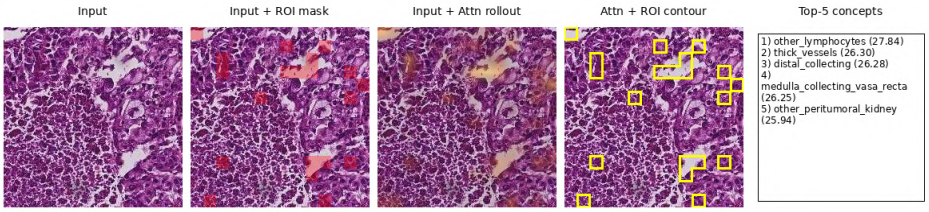

Fig. 27. PLIP concept grounding example for iBOT (T=NOT_TUMOR, P=pRCC). Top concepts in panel (e): lymphocytes (C024), thick vessels (C025), distal collecting (C052), medulla collecting vasa recta (C022), peritumoral kidney (C021). Despite non-tumor/context-like PLIP concepts, the model predicts a tumor subtype (failure mode).

D. I-JEPA

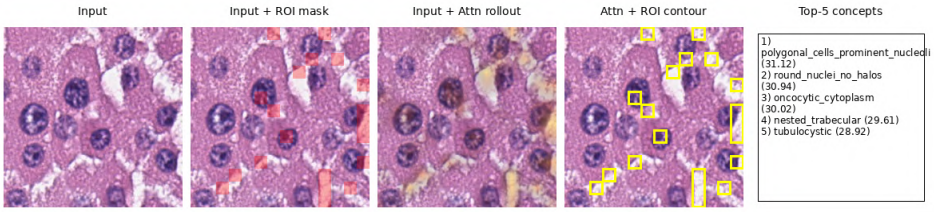

Fig. 28. PLIP concept grounding example for I-JEPA (T=ONCO, P=ONCO). Top concepts in panel (e): polygonal cells prominent nucleoli (C041), round nuclei no halos (C036), oncocytic cytoplasm (C040), nested trabecular (C038), tubulocystic (C042). PLIP concepts align with oncocytoma morphology.